

Supplementary Information for Precision-Retention Frontiers for Low-Precision Analog Vision via Ensemble Hardware-Aware Training

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1 Supplementary Information

1.1 Array-Level Non-Ideality Schematics

The first two supplementary figures define the layout-level perturbations used in the array non-ideality sensitivity probes. Placing these schematics first keeps the figure numbering aligned with the appendix visual narrative.

1.2 Differential-Pair Asymmetry Sensitivity

Systematic asymmetry between the positive and negative branches of differential weight mapping is evaluated by injecting controlled mismatch into the canonical uniform-noise HAT model. Consistent with the implementation, the branch update includes an offset correction so that the imposed asymmetry does not add a trivial constant differential bias from the shared conductance floor:

$$G_{\text{eff}}^{(\alpha)} = (1 + \alpha)G^+ - (1 - \alpha)G^- - 2\alpha G_{\text{min}} \quad (1)$$

where $\alpha \in [0, 0.10]$ is the asymmetry factor. Table S1 summarizes the accuracy degradation.

Table S1: Differential-pair asymmetry sensitivity. Accuracy degradation under systematic branch mismatch for Tiny-ViT V4 (CIFAR-10).

Asymmetry α	Mean Accuracy	Std Dev	Degradation
0.0 (baseline)	91.78%	0.10%	0.00%
0.01 (1%)	90.29%	0.15%	1.49%
0.02 (2%)	85.73%	0.59%	6.05%
0.05 (5%)	33.81%	1.76%	57.97%
0.10 (10%)	10.00%	0.00%	81.78%

The results indicate that differential-pair symmetry is an important layout constraint: asymmetry up to 1% is well-tolerated, but performance degrades nonlinearly beyond 2%, with complete failure at 10%. This motivates the need for explicit mismatch calibration or resistance balancing circuits in fabricated organic CIM arrays.

1.3 Physical Non-Ideality Sensitivity

Position-dependent IR drop and sneak-path currents are evaluated by injecting controlled non-idealities into the canonical uniform-noise HAT model. The two perturbations are implemented independently at the conductance level:

$$G^{(\text{IR})} = G(1 - \delta_{\text{IR}}), \quad \delta_{\text{IR}} \sim \mathcal{U}(0, \delta_{\text{IR}}^{\text{max}}) \quad (2)$$

$$G^{(\text{sneak})} = G + \epsilon_{\text{sneak}}, \quad \epsilon_{\text{sneak}} \sim \mathcal{N}(0, \sigma_{\text{sneak}} G_{\text{max}}) \quad (3)$$

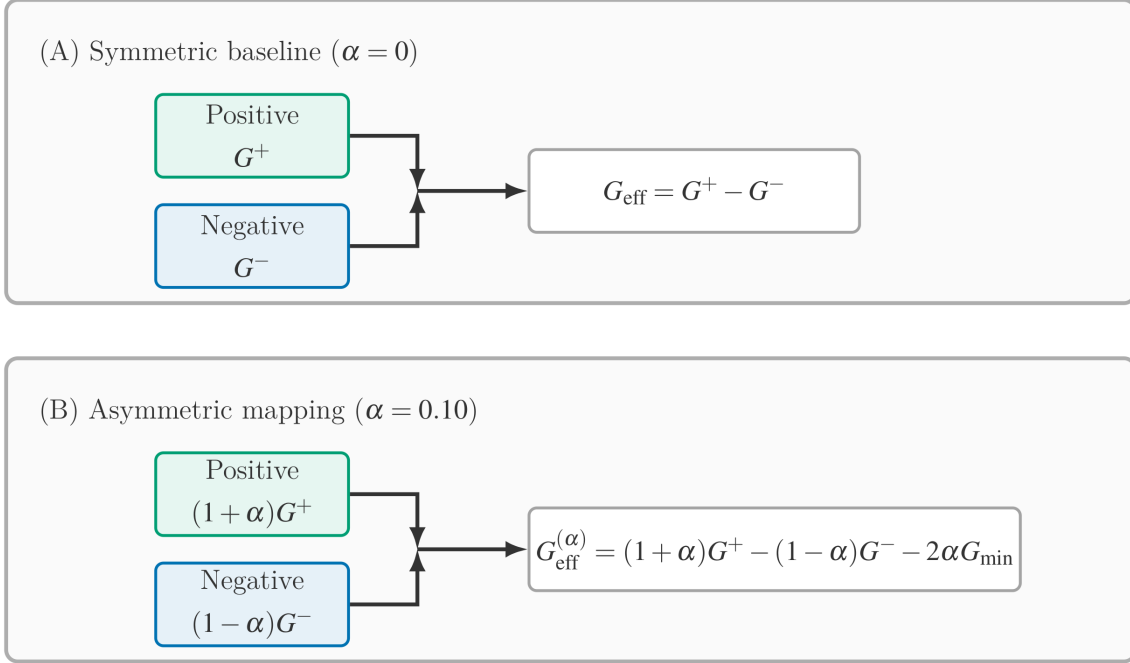


Figure S1: Differential-pair asymmetry concept. Systematic branch mismatch changes the effective differential signal; Table S1 gives the accuracy sensitivity.

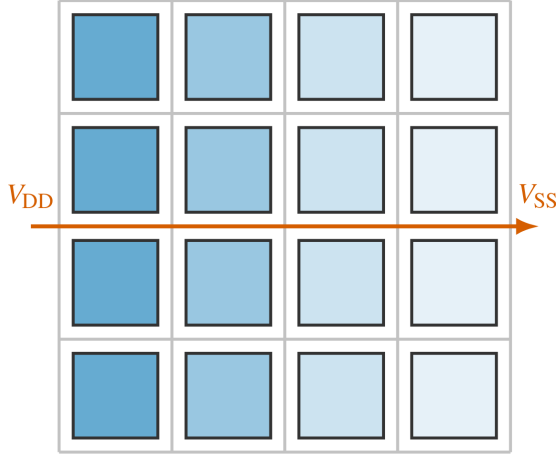
where $\delta_{\text{IR}}^{\text{max}} \in [0, 0.03]$ and $\sigma_{\text{sneak}} \in [0, 0.02]$ are the sweep controls based on ReRAM literature estimates. Table S2 summarizes the accuracy degradation.

Table S2: Physical non-ideality sensitivity. Accuracy degradation under IR drop and sneak-path effects for Tiny-ViT V4 (CIFAR-10).

IR Drop	Sneak Path	Mean Accuracy	Degradation
0.0%	0.0%	91.86%	0.00%
0.0%	1.0%	90.90%	0.96%
0.0%	2.0%	89.36%	2.50%
1.0%	0.0%	91.45%	0.41%
1.0%	1.0%	91.31%	0.55%
1.0%	2.0%	90.03%	1.83%
2.0%	0.0%	91.29%	0.57%
2.0%	1.0%	91.21%	0.65%
2.0%	2.0%	89.88%	1.98%
3.0%	0.0%	91.12%	0.74%
3.0%	1.0%	91.11%	0.75%
3.0%	2.0%	89.70%	2.16%

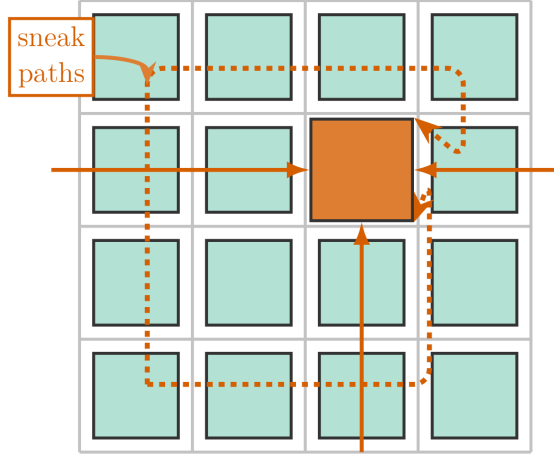
The results demonstrate that the hybrid-analog architecture maintains reasonable accuracy even under combined non-idealities. The worst-case degradation (2.16% at 3% IR + 2% sneak) remains limited within the tested lower-bound probe range and therefore provides a sensitivity check consistent with the use of first-order behavioral models for early-stage design space exploration, rather than a full parasitic-validation result for scaled organic arrays.

(A) IR drop
position-dependent
voltage loss



effective conductance
decreases along wordlines

(B) Sneak paths
unselected-cell
current leakage



leakage adds
readout error

$$G_{\text{read}} = G_{\text{ideal}} + \Delta G_{\text{array}}; \text{ Table ?? reports the tested error range.}$$

Figure S2: Physical non-idealities in resistive arrays. IR drop and sneak paths perturb the read conductance; Table S2 reports the first-order sensitivity probe.

1.4 Extended Methods

This section provides additional technical details on the system-level modeling and the physical simulation pipeline. We first detail the architectural partitioning between analog and digital domains, followed by the specific weight-to-conductance mapping procedures and the mathematical surrogates used for hardware-aware training.

1.4.1 Analog/Digital Operator Mapping

The hybrid CIM architecture partitions operators based on their computational density and weight reuse. Table S3 provides the full operator-level mapping rationale for the hybrid deployment used in our benchmarks.

1.4.2 Weight-to-Conductance Mapping Pipeline

The conversion of floating-point neural weights into physical device states is a multi-step process that accounts for both architectural quantization and physical device variability. Digital weights are first normalized and quantized, then decomposed into differential conductance pairs, perturbed by the stochastic device profile, and finally recovered via differential readout during inference.

Table S3: Analog/digital operator mapping rationale.

Operator	Domain	Rationale
Patch embedding (conv)	Analog	High weight reuse, regular access pattern; natural fit for crossbar MVM
QKV projections	Analog	Dense matrix-vector multiplications; weights stationary during inference
Attention output proj.	Analog	Post-attention linear projection; same characteristics as QKV
Feed-forward networks	Analog	MLP layers dominate parameter count; efficient crossbar utilization
Depthwise convolutions	Digital	Low weight reuse, irregular spatial access; poor crossbar utilization ¹
Layer normalization	Digital	Requires mean/variance computation across features; hard to parallelize on crossbar
$\mathbf{QK}^\top$ (attention scores)	Digital	Dynamic, input-dependent computation; irregular memory access ²
Softmax	Digital	Non-linear transcendental function; precision-critical; retained in the digital path in contemporary hybrid CIM ViT mappings ^{2;3}

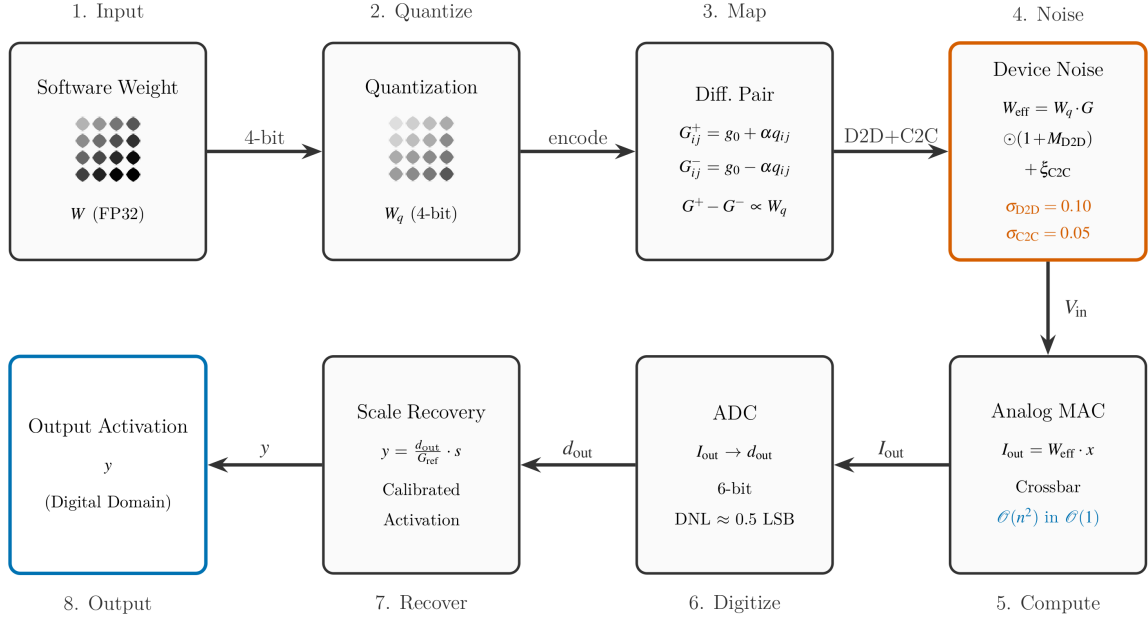


Figure S3: Weight-to-conductance mapping pipeline. Digital weights are quantized, mapped to differential conductance pairs, perturbed by D2D/C2C device terms, accumulated in the analog domain, digitized, and recovered with a layer-wise scale factor.

1.4.3 Hybrid System Architecture

The overall hardware stack (Fig. S4) integrates the analog-CIM accelerator with a digital control path to handle the non-linear and dynamic components of the transformer architecture.

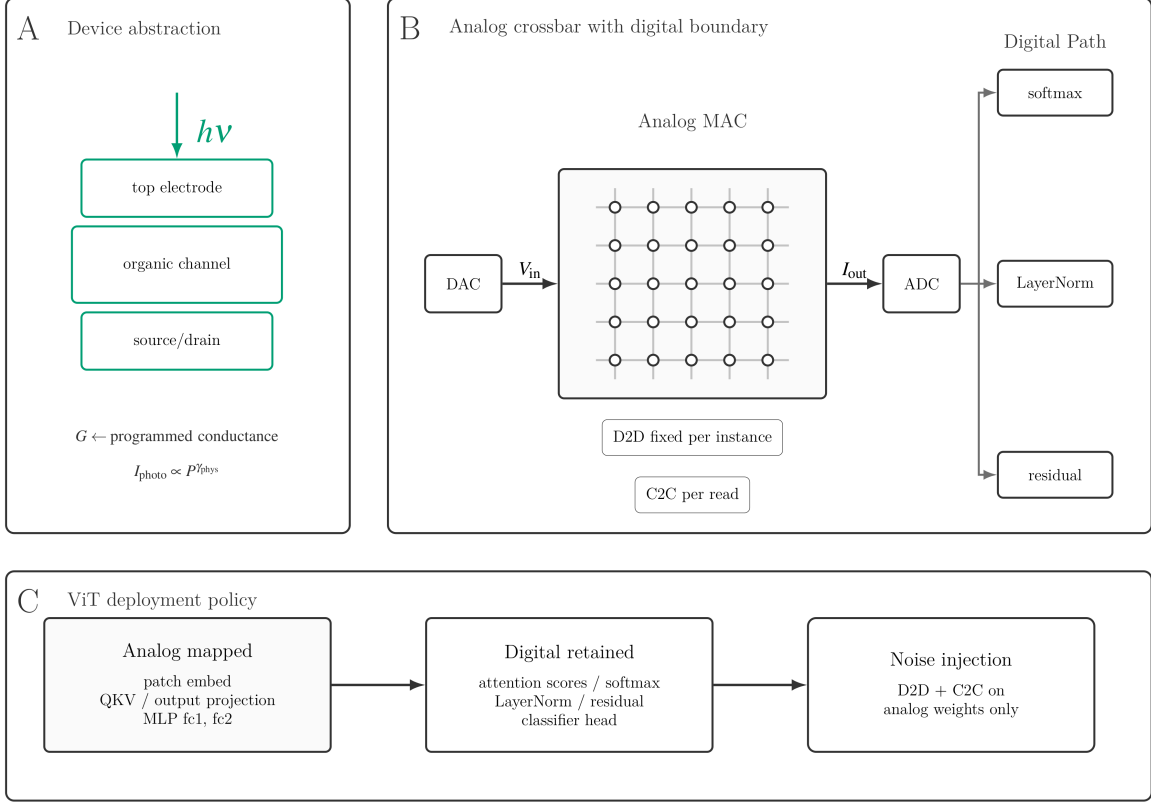


Figure S4: Hybrid analog-CIM inference stack. Dense weight-stationary operators are executed on analog crossbars, whereas dynamic attention and control-heavy operators remain in the digital path.

1.4.4 Nonlinear-Write Surrogate

To model the limitations of physical programming, we use a behavioral surrogate in the backward pass. In the surrogate model, nonlinear write modifies the STE backward pass according to the present conductance state:

$$\frac{\partial L}{\partial G} = \begin{cases} \frac{\partial L}{\partial \tilde{G}} \left(\frac{G_{\max} - G}{G_{\max} - G_{\min}} \right)^{NL_{LTP}-1}, & \frac{\partial L}{\partial \tilde{G}} \geq 0 \\ \frac{\partial L}{\partial \tilde{G}} \left(\frac{G - G_{\min}}{G_{\max} - G_{\min}} \right)^{|NL_{LTD}|-1}, & \frac{\partial L}{\partial \tilde{G}} < 0 \end{cases} \quad (4)$$

The absence of an explicit NL prefactor in Eq. 4 is intentional: the surrogate scales the STE by the normalized conductance position, modeling state-dependent update difficulty (weaker updates near the bounds), rather than by the derivative of a power-law function $f(G) = G^{NL}$. This is therefore a behavioral proxy for position-dependent gradient attenuation, not a strict functional derivative. When `use_second_order_ste=True`, a second-order Taylor correction adds a negative-signed brake to both branches, preventing aggressive bound-pushing without changing the first-order scaling.¹ Ideal write is recovered when $NL_{LTP} = 1$ and $|NL_{LTD}| = 1$.

¹Branch A ratified in canonical commit `ab56c2d`.

1.5 Mathematical Formalism of Hardware-Aware Training

We define the optimization objectives for the various training regimes studied. Standard hardware-aware training (HAT) optimizes weights θ against a single, fixed device-to-device (D2D) mismatch field M_0 :

$$\theta_{\text{std}}^* = \arg \min_{\theta} \frac{1}{N} \sum_{n=1}^N \ell(f(x_n; \theta, M_0), y_n), \quad (5)$$

Ensemble HAT replaces the fixed mask with samples from the D2D distribution during every training epoch:

$$\theta_{\text{ens}}^* = \arg \min_{\theta} \mathbb{E}_{M \sim p(M)} \left[\frac{1}{N} \sum_{n=1}^N \ell(f(x_n; \theta, M), y_n) \right]. \quad (6)$$

Fresh-instance transfer is then evaluated over unseen arrays M' drawn from the same distribution to assess generalizability:

$$\text{Acc}_{\text{fresh}} = \mathbb{E}_{M' \sim p(M)} [\text{Acc}(f(\cdot; \theta_{\text{ens}}^*, M'))]. \quad (7)$$

1.6 Physical Front-end Modeling

The physical transduction model accounts for the non-linear response and noise characteristics of the organic phototransistor front end. The organic phototransistor front end is modeled with an inverse-gamma preprocessor P_{in} applied to the normalized input $X \in [0, 1]$:

$$P_{\text{in}} = X^{1/\gamma_{\text{phys}}}. \quad (8)$$

The photocurrent I_{photo} subsequently incorporates physical transduction nonlinearity and shot noise ϵ_{shot} :

$$I_{\text{photo}} = \alpha P_{\text{in}}^{\gamma_{\text{phys}}} + I_{\text{dark}} + \epsilon_{\text{shot}} \approx \alpha X + I_{\text{dark}} + \epsilon_{\text{shot}}, \quad (9)$$

where $\text{Var}[\epsilon_{\text{shot}}] \propto I_{\text{photo}}$. The signal is renormalized per sample before being passed to the first neural layer:

$$X_{\text{norm}} = \frac{I_{\text{photo}} - I_{\text{min}}}{I_{\text{max}} - I_{\text{min}} + \epsilon}. \quad (10)$$

1.7 Supplementary Note S-Notation: Model Configurations

Table S4 defines the model configurations referenced throughout this study. These designations unify the experimental variants across different backbones and non-ideality regimes.

Table S4: Unified model notation.

Notation	Physical Regime	Training Protocol
Digital	Ideal software baseline	FP32 or quantized digital training
Fixed-mask	Stochastic noisy crossbar	Fixed-instance hardware-aware training
Ensemble HAT	Stochastic noisy crossbar	Per-epoch D2D mask resampling (Ours)
PCM 8-bit	Realistic PCM UnitCell	AnalogSGD (100 epochs)
PCM 6-bit	Realistic PCM UnitCell	AnalogSGD (100 epochs)
PCM 4-bit	Realistic PCM UnitCell	AnalogSGD (100 epochs)

1.8 Detailed Experiment Matrix

To provide a comprehensive view of the experimental coverage, we detail the specific model variants and physical regimes evaluated across our three primary backbones (ResNet, ConvNeXt, and Tiny-ViT). The experiments span a spectrum from ideal digital baselines to extreme physical stress tests involving high mismatch and non-linear programming. For clarity and brevity, these regimes are denoted V1–V8 throughout the Supplementary Information, mirroring the architecture-specific prefixes used in the main text (e.g., C for ConvNeXt, R for ResNet). V3 serves as our baseline noisy deployment under fixed D2D masking, while V7–V8 represent exploratory studies on long-term retention drift that were used to define the boundaries of the precision-retention frontier.

Four physical-extension studies probe richer device regimes on CIFAR-10:

- Proportional-noise HAT (V4_proportional_HAT) repeats V4 for 100 epochs with state-dependent proportional noise and uses 10-run Monte Carlo sweeps under both proportional and uniform-noise semantics.
- Nonlinear-write HAT (V4_NL2_HAT) repeats V4 for 100 epochs with nonlinear-write dynamics active during STE gradient computation ($NL_{LTP} = +2.0$, $NL_{LTD} = -2.0$) and uses the same 10-run Monte Carlo evaluation.
- Ensemble HAT (V4_Ensemble_HAT) keeps the canonical V4 recipe but resamples the fixed D2D realization at each epoch; transferability is then reported on 10 fresh hardware instances with 5 Monte Carlo evaluations per instance.
- ConvNeXt proportional-noise HAT (C4) applies the same proportional-noise HAT protocol to ConvNeXt-Tiny for 200 epochs as a cross-architecture comparison.

These runs are treated as physical-extension studies rather than as part of the grouped cross-dataset comparisons in the main text.

Table S5: Digital FP32 baselines for the evaluated backbones and datasets.

Dataset	ResNet-18	ConvNeXt-Tiny	Tiny-ViT-5M
CIFAR-10	95.46%	90.74%	98.06%
CIFAR-100	78.64%	64.12%	86.94%

Table S6: Summary of classification accuracy (%) across architectures and physical regimes. Cross-dataset deployment rows report locked best-checkpoint values; entries with $\pm$ denote stochastic physical-extension evaluations. n.e. denotes a protocol not evaluated, not a failed run.

Architecture	Regime	CIFAR-10	CIFAR-100
ResNet-18	FP32 Digital	95.46	78.64
ResNet-18	Standard noisy deployment (R3)	16.48	1.00
ResNet-18	Uniform-noise HAT (R4)	90.37	1.00
Tiny-ViT-5M	FP32 Digital	98.06	86.94
ConvNeXt-Tiny	FP32 Digital	90.74	64.12
ConvNeXt-Tiny	Standard noisy deployment (C3)	70.48	23.86
ConvNeXt-Tiny	Uniform-noise HAT (C4)	89.91	60.54
ConvNeXt-Tiny	Proportional-noise HAT (3-seed extension)	n.e.	84.75 ± 0.72
Tiny-ViT-5M	Standard noisy deployment (V3)	89.54	44.06
Tiny-ViT-5M	Uniform-noise HAT (V4)	91.94	65.48
Tiny-ViT-5M	Front-end compensation (V6)	82.58 (single-seed)	n.e.

1.9 Parameter Provenance and Sensitivity

To establish the robustness of our system-level conclusions, we provide a detailed audit of the parameter provenance and the associated sensitivity checks. This section explicitly distinguishes between values directly reported in device literature, literature-derived proxy estimates, and analytical assumptions where direct measurements are currently unavailable.

1.9.1 Three-Seed Summary for the Canonical V4 Path

To document the reproducibility of the canonical Tiny-ViT Ensemble HAT path, we report the protocol-matched three-seed fresh-instance summary directly in the Supplementary Information. Each seed was evaluated on 10 fresh D2D instances with 5 Monte Carlo passes per instance, and the final cross-seed statistic is computed from the three seed means.

Table S7: Three-seed fresh-instance reproducibility summary for the uniform-noise Ensemble HAT regime (V4).

Seed / Aggregate	Fresh Accuracy	Statistic Type
123	$86.37 \pm 1.54\%$	10 fresh instances $\times$ 5 MC
456	$86.12 \pm 0.72\%$	10 fresh instances $\times$ 5 MC
789	$85.99 \pm 1.94\%$	10 fresh instances $\times$ 5 MC
Cross-seed aggregate	$86.16 \pm 0.19\%$	mean $\pm$ sample std of seed means
Pooled fresh instances	$86.16 \pm 1.52\%$	mean $\pm$ sample std over 30 instance means

Table S8: Parameter provenance tracking matrix.

Parameter	Canonical Organic Profile	Zhang 2025 OPECT Case Study	Nonlinear-Write Stress Tests	Provenance / Derivation Notes
Conductance Range ($G_{\max}/G_{\min}$)	10	47.3	10	Canonical anchored to scalable OPECTs; Zhang derives from reported max/min current levels ⁴ .
Effective States (n_{states})	16 (4-bit)	34	16 (4-bit)	Canonical matches general low-precision target. Zhang anchored to Fig.3h & Supp.Fig.8 ⁴ .
Cycle-to-Cycle Noise (σ_{C2C})	5%	2%	5% (Proportional)	Zhang value is a proxy estimate from 8-cycle repeatability (Supp.Fig.15) ⁴ . Proportional stress scales $\sigma \propto G $.
Device-to-Device Mismatch (σ_{D2D})	10%	3%	10% (Proportional)	Zhang uses 3% as a conservative conductance-domain proxy from the reported $\sim 1\%$ V_{th} spread ⁴ .
Retention Decay (τ_1, τ_2, A_0)	140ms, 610ms, 0.6	Not fitted	140ms, 610ms, 0.6	Canonical dual-exponential fit anchored to Vincze ⁵ . Zhang (Fig.2d) is qualitative only.
Nonlinear Write Asymmetry (NL)	1.0 (LTP), -1.0 (LTD)	Canonical defaults retained	2.0 (LTP), -2.0 (LTD)	Zhang does not report a pulse-level NL fit, so the case study keeps the canonical symmetric-write defaults rather than injecting a separate literature-fit NL term. Stress tests ($NL = 2$) enforce severe gradient-scaling asymmetry.
Noise Mode	Uniform	Uniform	Proportional	Uniform injects noise normalized to full range. Proportional applies state-dependent noise magnitudes.

1.9.2 Vincze et al. 2025 Parameter Extraction

Because Vincze et al.⁵ is currently available as a 2025 Early Access paper, we list the extracted parameters used to define the canonical organic profile explicitly here:

- Conductance Window: $G_{\max}/G_{\min} \approx 10$, anchored to the scalable operation regime.
- Retention Time Constants: The dual-exponential fit for temporal drift yields a fast decay constant $\tau_1 = 140$ ms, a slow decay constant $\tau_2 = 610$ ms, and an initial amplitude ratio $A_0 = 0.6$. These align with the short-term and long-term plasticity dynamics reported for the device.
- Nonlinear Write Asymmetry (NL): While the canonical profile uses symmetric writes ($NL = 1.0$), the physical profile derived from the literature indicates moderate write non-linearity corresponding to $NL_{\text{LTP}} = 2.0$ and $NL_{\text{LTD}} = -2.0$. In our simulations, $NL = 2.0$ maps to a conductance saturation asymmetry (LTP vs. LTD magnitude at the window midpoint) of approximately 3:1, a regime consistent with reported long-pulse dynamics in DNTT-based organic synaptic transistors⁵. We evaluate this directly in our severe stress tests.

1.9.3 Proxy Estimate Sensitivity Analysis

The Zhang 2025 OPECT values for σ_{C2C} and σ_{D2D} are transparent proxy estimates rather than directly measured full-array noise distributions. We therefore ran a compact 4×5 sensitivity sweep on the Ensemble HAT model, spanning $\sigma_{\text{C2C}} \in \{1, 2, 5, 8\}\%$ and $\sigma_{\text{D2D}} \in \{2, 3, 5, 10, 15\}\%$ while keeping the remaining case-study settings fixed. Figure S5 shows that the conclusion is insensitive to the exact proxy choice: performance changes primarily along the D2D axis, whereas the tested C2C range remains effectively masked by the digital recovery path.

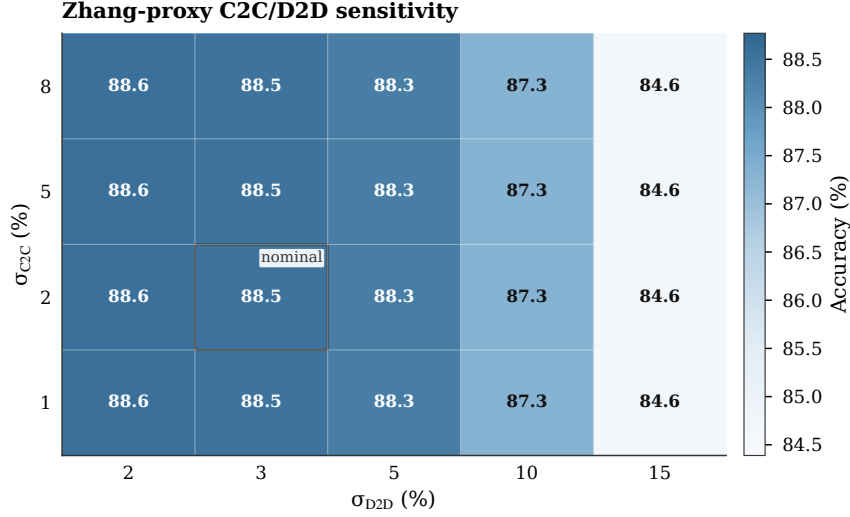


Figure S5: Zhang-proxy C2C/D2D sensitivity sweep. Cells show 10-MC mean accuracy; the nominal proxy point is outlined. Accuracy is driven mainly by D2D mismatch, while C2C is flat within MC uncertainty.

Table S9: Ensemble HAT accuracy under a Zhang-proxy C2C/D2D sensitivity sweep.

C2C \ D2D	2%	3% (Nominal)	5%	10%	15%
1%	88.57%	88.53%	88.33%	87.30%	84.59%
2% (Nominal)	88.57%	88.53%	88.33%	87.30%	84.59%
5%	88.57%	88.53%	88.33%	87.30%	84.59%
8%	88.57%	88.53%	88.33%	87.30%	84.59%

95% confidence intervals are sub-0.1 pp for all nominal points.

Table S10: Hook-based ADC non-ideality check for the canonical Tiny-ViT V4 deployment. Offset, gain, and INL are injected at calibrated analog-layer outputs during full-test inference on CIFAR-10.

Configuration	Mean Acc. (%)	Std. (%)	Δ vs. ideal (pp)
Ideal	82.04	0.16	+0.00
Offset ± 0.5 LSB	82.07	0.21	+0.03
Gain $\pm 5\%$	81.87	0.30	-0.17
INL 0.5 LSB	80.85	0.12	-1.19
Combined realistic	81.86	0.28	-0.18
Combined pessimistic	76.90	0.27	-5.14

Moderate ADC non-idealities change accuracy by <0.2 pp, whereas the pessimistic combination degrades by 5.14 pp, confirming that peripheral precision is gate-keeping only once errors exceed the moderate range.

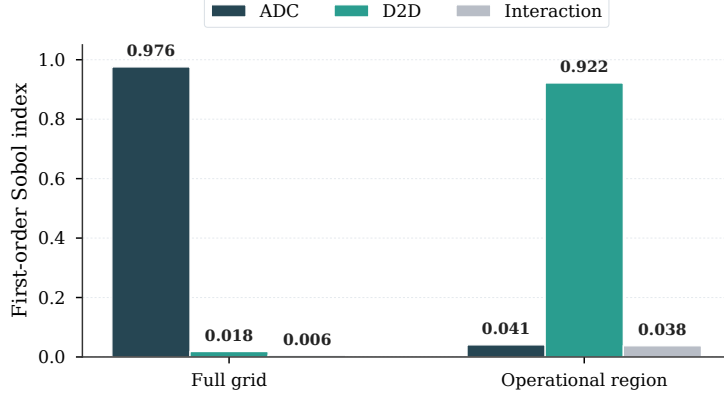


Figure S6: First-order Sobol sensitivity indices. The full D2D–ADC grid is dominated by ADC precision; after restricting to $\text{ADC} \geq 6$ bits and $\sigma_{\text{D2D}} \leq 15\%$, residual variance is dominated by D2D mismatch.

1.9.4 Uniform vs. State-Dependent Retention Comparison

We compared uniform retention against a state-dependent variant in which high-conductance states decay 20% faster, using the retained Ensemble-HAT CIFAR-10 model as the test case. This comparison is meant to check whether the simpler uniform-retention assumption changes the measured accuracy trajectory in this setting, not to rule out state dependence in other devices or workloads.

Table S11: Retention accuracy comparison: uniform vs. state-dependent decay.

Time (s)	Uniform (%)	State-Dep (%)	Diff (pp)
0	90.77	90.80	+0.03
1	90.07	90.08	+0.01
10	89.78	89.81	+0.03
100	89.87	89.89	+0.02
1000	89.82	89.90	+0.08

Across the tested retention times, the two models differ by less than 0.1 percentage points. This supports the uniform-retention approximation for the retained Ensemble-HAT CIFAR-10 diagnostic, while leaving open the possibility that stronger conductance dependence could matter in other device profiles.

Table S12: Classification accuracy (%) on Flowers-102 across physical regimes. The asterisk marks a single-run ConvNeXt boundary estimate and should not be interpreted as a stable cross-architecture baseline.

Architecture	Regime	Flowers-102
Tiny-ViT-5M	FP32 Digital	97.97
ConvNeXt-Tiny	FP32 Digital	33.22 [†]
Tiny-ViT-5M	V3 (Canonical Noise)	4.81 ± 1.25
Tiny-ViT-5M	V4 (Canonical HAT)	22.48 ± 1.82

Because the ConvNeXt Flowers-102 baseline is available only as a single-run estimate, we use this dataset primarily as a low-data stress test rather than as a definitive cross-architecture comparison point.

1.10 Supplementary Methods

1.10.1 Differential Conductance Mapping Details

Each analog-mapped weight tensor is converted into a differential-pair conductance representation over a bounded physical range $[G_{\min}, G_{\max}]$. The positive and negative branches are quantized to n_{states} levels with a straight-through estimator (STE), and the effective analog weight is recovered from the differential conductance. The layer-wise digital scale recovery is

$$\hat{W}_{ij} = s_{\ell} G_{\text{eff},ij}, \quad s_{\ell} = \frac{w_{\max}}{\Delta G}, \quad \Delta G = G_{\max} - G_{\min},$$

with the optional retention-time recalibration

$$s_{\ell}^{\text{ret}} = \frac{w_{\max}}{\max_{i,j} |G_{\text{eff},ij}| + \varepsilon}.$$

The recovered weight step is therefore

$$\Delta_{\hat{W}} = s_{\ell} \frac{\Delta G}{n_{\text{states}} - 1},$$

so the scale-masking regime discussed in the main text corresponds to perturbations satisfying $|s_{\ell} \delta G_{\text{eff},ij}| < \Delta_{\hat{W}}/2$. We model this scale recovery as ideal calibrated digital post-processing rather than as an explicit physical peripheral circuit. Deterministic positive/negative branch asymmetry is not separately parameterized beyond the sampled D2D/C2C perturbations; any perfect common-mode cancellation implied by the differential form is a simplifying layout-level assumption.

1.10.2 Variability and Write Dynamics Modeling

Device-to-device (D2D) mismatch is modeled as a fixed Gaussian perturbation sampled once at initialization,

$$\delta g_{\text{D2D}} \sim \mathcal{N}(0, \sigma_{\text{D2D}} \Delta G), \quad \Delta G = G_{\max} - G_{\min},$$

whereas cycle-to-cycle (C2C) variation is resampled on each forward pass as

$$\delta g_{\text{C2C}} \sim \mathcal{N}(0, \sigma_{\text{C2C}} \Delta G).$$

In proportional mode, the same Gaussian law is used but the perturbation magnitude scales with the instantaneous conductance state, i.e. $\delta g \propto |G|$ rather than ΔG .

Nonlinear write modeling utilizes NL_LTP and NL_LTD coefficients to scale the STE backward pass. The surrogate follows

$$\frac{\partial L}{\partial G} = \begin{cases} \frac{\partial L}{\partial \tilde{G}} \left(\frac{G_{\max} - G}{G_{\max} - G_{\min}} \right)^{NL_{\text{LTP}} - 1}, & \frac{\partial L}{\partial \tilde{G}} \geq 0 \\ \frac{\partial L}{\partial \tilde{G}} \left(\frac{G - G_{\min}}{G_{\max} - G_{\min}} \right)^{|NL_{\text{LTD}}| - 1}, & \frac{\partial L}{\partial \tilde{G}} < 0 \end{cases}$$

so $NL_{\text{LTP}} = 1$ and $|NL_{\text{LTD}}| = 1$ recover the identity STE. Retention is implemented as a decay factor applied to the programmed conductance above the floor $G_{\min}$:

$$G(t) = G_{\min} + (G_{\text{prog}} - G_{\min}) f_{\text{ret}}(t),$$

with

$$f_{\text{ret}}(t) = A_0 + A_1 e^{-t/\tau_1} + A_2 e^{-t/\tau_2}, \quad A_1 = A_2 = \frac{1 - A_0}{2},$$

and parameters anchored to measured DNTT transients⁵. In the optional state-dependent extension, the same form is used with conductance-dependent effective time constants so that high-conductance states decay modestly faster.

1.10.3 Energy Profiler Implementation

The analytical energy model estimates cost across four categories:

1. Analog MAC: $E_{\text{MAC}} = N_{\text{ops}} \cdot E_{\text{cell}}$, which sets $E_{\text{cell}} = 100$ fJ per differential-pair MAC in the default profiler.
2. Data Conversion: ADC energy is charged per output element and DAC energy per input element, with behavioral DNL/INL modeling. The default implementation uses $E_{\text{ADC},8b} = 25$ fJ, $E_{\text{DAC},8b} = 30$ fJ, and $t_{\text{ADC},8b} = 100$ ns for a SAR-like 8-bit readout proxy.
3. Digital Operations: Fixed-point costs for Softmax, LayerNorm, and scaling.
4. Memory Traffic: SRAM read/write energy for weight programming and intermediate activation buffering.

The reported total is accumulated as

$$E_{\text{total}} = E_{\text{MAC}} + E_{\text{ADC}} + E_{\text{DAC}} + E_{\text{digital}} + E_{\text{buffer}},$$

with

$$E_{\text{ADC}} = N_{\text{ADC}} E_{\text{ADC},8b}, \quad E_{\text{DAC}} = N_{\text{DAC}} E_{\text{DAC},8b},$$

and the digital/buffer terms evaluated from operator and access counts. These constants are literature-derived proxy values for a first-order edge-node model, not measured circuit values, and any reported energy ratios are contingent on these assumptions. Interconnect overhead is behaviorally absorbed into the memory-access terms.

Supplementary Note S-Energy: Provenance of Energy Model Constants

This note documents the origin and selection criteria for every physical and architectural constant used in the first-order analytical energy profiler (§3.7 of the main text). Given the early stage of organic CIM hardware development, we utilize a combination of literature-anchored device metrics and established CMOS benchmarks to provide a comparative framework. The purpose is to allow reviewers to distinguish measured results in literature from order-of-magnitude engineering estimates used for higher-level system scaling.

S-Energy.1 Parameter table

S-Energy.2 Detailed provenance notes

Analog MAC (100 fJ). Gebregiorgis et al.⁶ report a three-terminal organic device array that achieves 100 fJ per MAC operation, decomposed as 70 fJ (device) + 20 fJ (readout) + 10 fJ (digital overhead). We adopt this number as a literature proxy: it is a measured result for a specific organic thin-film technology, but it is not guaranteed to match the conductance window, read voltage, or peripheral architecture of the organic devices targeted by the present framework. The conservative alternative (150 fJ) is the median reported for RRAM-based CIM accelerators and is used only in sensitivity sweeps.

8-bit ADC (25 fJ/conv). The value is a rounded median of the 8-bit SAR-ADC range reported in recent CIM literature (10–30 fJ/conv). Individual data points include Yoon et al. (24.8 fJ, TCAS-II 2025) and Zhang et al. (10.2 fJ, ISCAS 2022). Because these papers target different technology nodes, topologies, and sharing factors, we do not cite a single source; instead we label the 25 fJ value as an order-of-magnitude estimate suitable for first-order relative comparison.

Parameter	Value	Provenance	Status
$E_{\text{analog,MAC}}$	100 fJ	Gebregiorgis et al. (2023) ⁶	Literature proxy
$E_{\text{conservative}}^{\text{analog,MAC}}$	150 fJ	RRAM/CIM literature median	Order-of-magnitude estimate
$E_{\text{ADC,8b}}$	25 fJ/conv	CIM SAR-ADC literature median (10–30 fJ)	Order-of-magnitude estimate
$E_{\text{DAC,8b}}$	30 fJ/conv	Li et al. (2022) TCAS-I estimate (not in present bibliography)	Order-of-magnitude estimate
$E_{\text{digital,INT8}}$	0.4 pJ	Horowitz (2014) ⁷ + 28 nm scaling	Literature-derived
$E_{\text{digital,FP32}}$	2.5 pJ	Horowitz (2014) ⁷ + 28 nm scaling	Literature-derived
$E_{\text{SRAM,read}}$	5 pJ	Horowitz (2014) ⁷	Literature-derived
$E_{\text{DRAM,read}}$	1300 pJ	Horowitz (2014) ⁷	Literature-derived
E_{softmax}	15 pJ/elem	Analytical estimate (exp + divide)	Engineering estimate
$E_{\text{layernorm}}$	8 pJ/elem	Analytical estimate (sqrt + divide)	Engineering estimate

Table S13: Provenance of energy-model constants. Literature proxies and order-of-magnitude estimates are used for relative comparison, not routed-circuit prediction.

8-bit DAC (30 fJ/conv). Internal project notes reference Li et al. (2022, TCAS-I), who report a time-multiplexing CIM architecture with a full energy breakdown of 136 fJ/MAC, of which the DAC contribution is 10 fJ under heavy column sharing. The 30 fJ value used here is a higher, architecture-agnostic estimate that assumes less aggressive sharing. Because the TCAS-I paper is not included in the present bibliography, the 30 fJ value is labeled as an order-of-magnitude estimate rather than a direct citation.

Digital CMOS (0.4 pJ INT8, 2.5 pJ FP32). Both numbers are derived from Horowitz (2014)⁷, the most widely used CMOS energy benchmark in the CIM community. The raw 45 nm values (FP32 $\approx$ 4.6 pJ, INT32 $\approx$ 3.2 pJ) are scaled to 28 nm using standard quadratic voltage-and-capacitance scaling, yielding 2.5 pJ and 0.4 pJ respectively. These are literature-derived estimates.

S-Energy.3 Limitations and intended use

All constants in Table S13 are analytical placeholders. They are intended for:

1. Relative comparison between analog and digital execution paths under identical operation-count assumptions;
2. Sensitivity analysis (e.g., varying routing overhead, ADC precision, or analog-MAC cost);
3. Design-space exploration before tape-out, not as silicon power predictions.

Absolute energy numbers (e.g., 273.94 μ J per inference, 11.45 $\times$ reduction) should be read as first-order upper-bound projections, not routed-circuit measurements. Supplementary sensitivity tables (S6–S7) show that the analog-digital ranking is robust to $\pm 30\%$ variation in any single parameter and to 10–50% unmodeled routing overhead.

1.11 Supplementary Figures

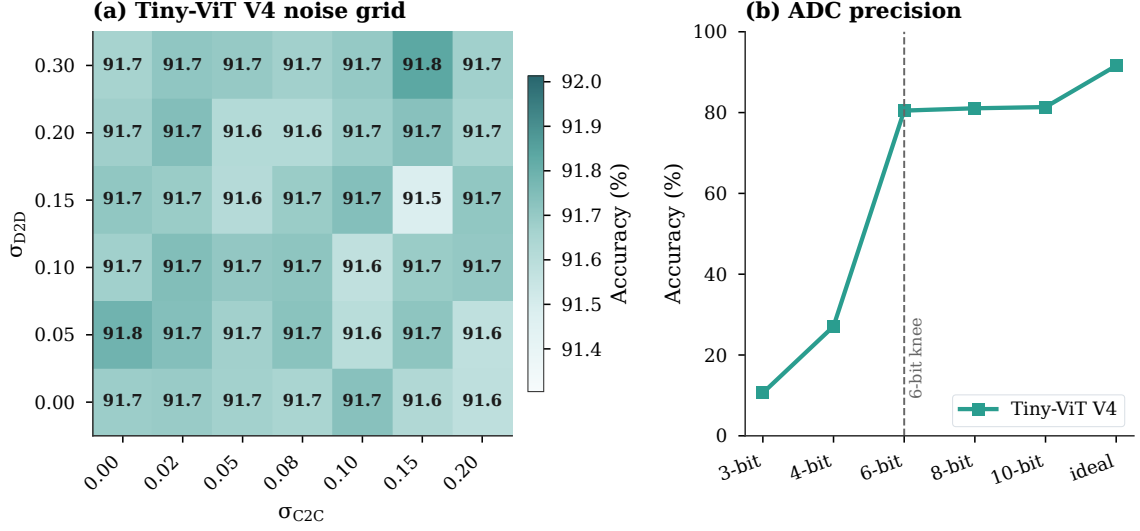


Figure S7: Continuous noise and ADC sensitivity. Left: mean accuracy over the joint $\sigma_{C2C}/\sigma_{D2D}$ sweep. Right: ADC-bit sweep, showing the 6-bit knee and higher-bit saturation.

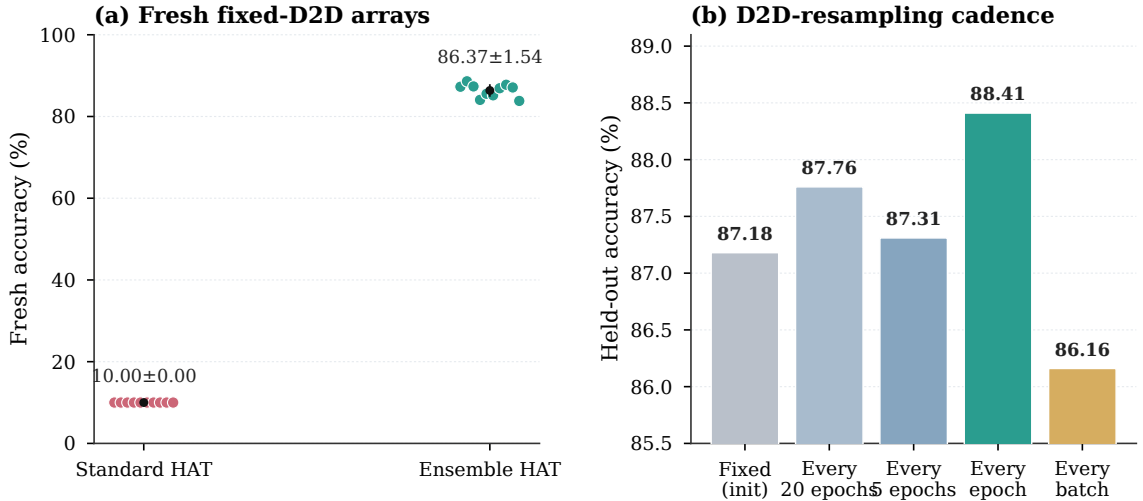


Figure S8: Fresh-instance robustness and resampling cadence. Left: Standard HAT collapses on fresh arrays, while Ensemble HAT transfers. Right: a 50-epoch scan shows epoch-level D2D resampling outperforming fixed-mask, slower-refresh, and short-run per-batch controls.

To complement the qualitative maps in Supplementary Fig. S9, we also tracked the mean head-averaged attention entropy for the three representative samples shown there. Averaged across the samples, the entropy rises from 3.38 in the digital baseline to 3.61 under standard noisy deployment, indicating a broader and less selective spatial focus. Under HAT, the entropy drops to 3.07, and the model recovers the correct class label on all three displayed examples, whereas the fixed-mask noisy model is correct on only one. These examples provide a compact quantitative complement to the attention maps by showing that HAT restores a sharper task-aligned focus pattern.

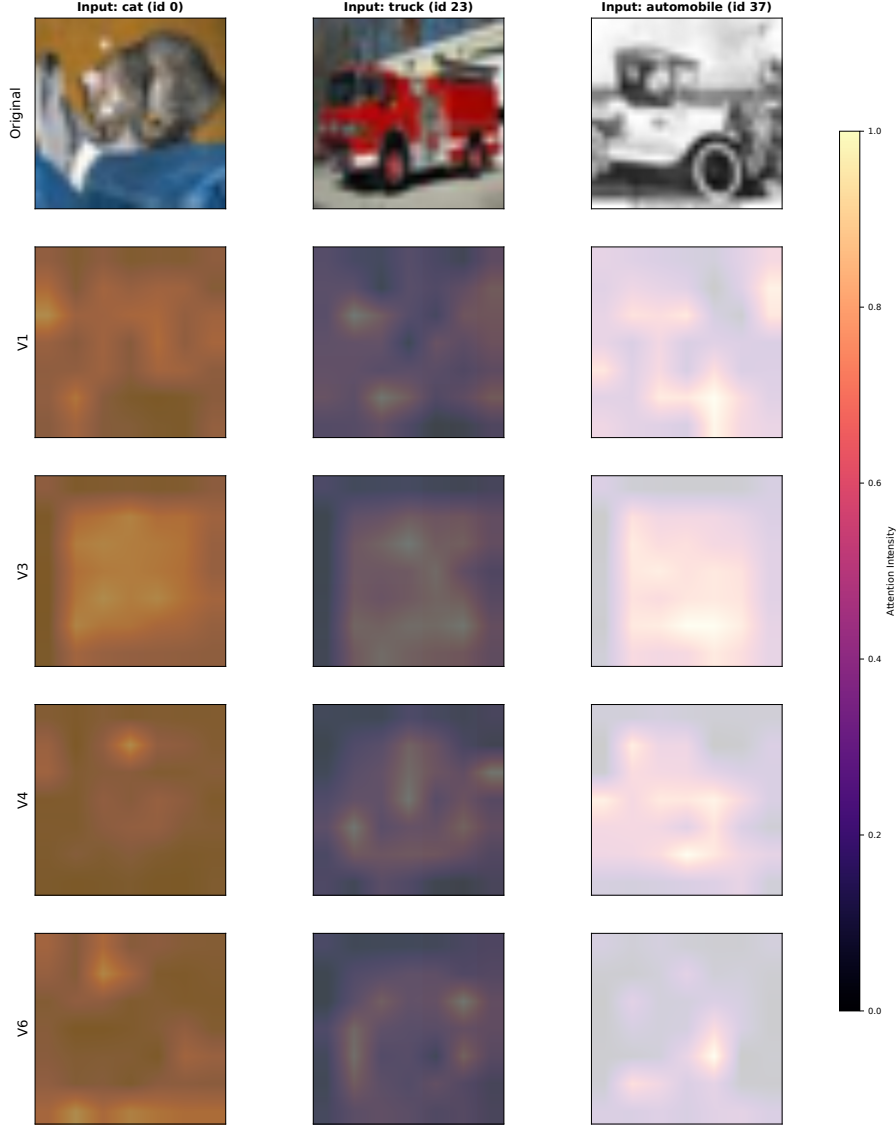


Figure S9: Attention heatmaps across deployment regimes. Columns show three CIFAR-10 samples; rows show input, digital reference, fixed-mask baseline, Ensemble HAT, and front-end compensation.

1.12 Extended Results: Retention and Frontend Distortion

1.12.1 Retention and Temporal Drift Details

Supplementary Figure S10 (Left) shows the corrected V4 retention decay with dynamic scale recalibration and co-decay of the retained D2D buffers. The curve exhibits a two-phase behavior: a rapid early drop from 91.63% to 82.66% at 1 s, followed by a broad plateau near 79% accuracy from 10 s to 10,000 s (79.13%, 79.05%, 79.35%, and 79.51% at 10, 100, 1000, and 10,000 s, respectively). Under the present first-order retention model, this indicates that long-horizon analog inference remains partially viable once gain recalibration is included in the inference stack. The retention results reported here were obtained under the same uniform double-exponential decay model described in the main-text methodology section. The codebase now additionally supports true state-dependent retention where high-conductance states decay faster, but the canonical V4 results in this paper use the uniform model. A direct comparison between these two models (detailed in Section 1.9) shows that the accuracy difference remains below 0.1 percentage points across all tested time intervals, validating the sufficiency of the uniform model for the present parameter regime.

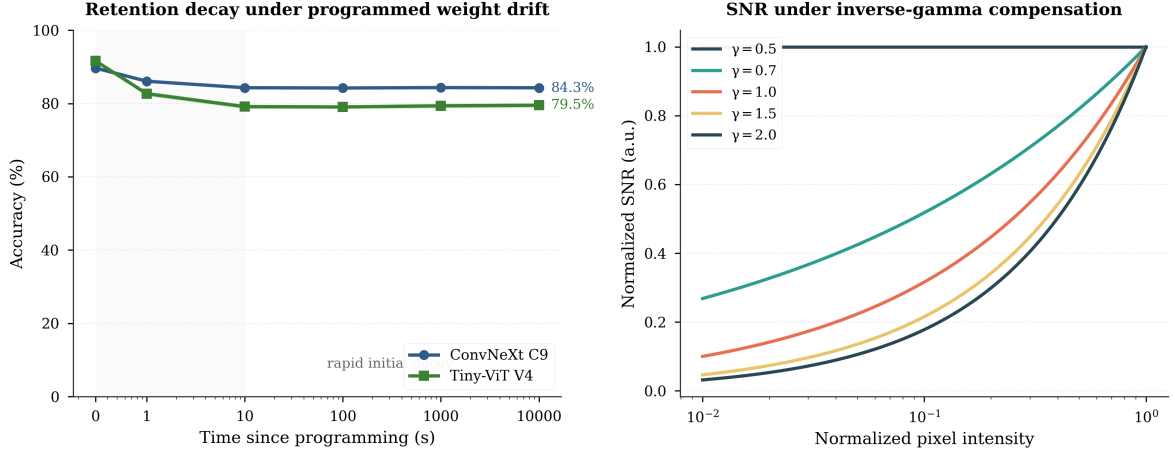


Figure S10: Extended physical performance bounds. Left: Retention decay under drift for ConvNeXt C9 and Tiny-ViT V4, showing an early drop followed by a broad plateau. Right: Analytical SNR trends under inverse-gamma compensation derived from the front-end response model.

1.12.2 Theoretical Analysis of Frontend Compensation

Distinction from ISP gamma correction. The inverse-gamma preprocessor $P_{\text{in}} = X^{1/\gamma_{\text{phys}}}$ is superficially similar to gamma correction in digital image processing (ISP), but the two serve different physical purposes and operate in different noise regimes. In ISP, gamma encoding ($\gamma_{\text{ISP}} \approx 2.2$) compresses high-intensity pixel values to match the nonlinear response of the human visual system, while its inverse (gamma decoding) expands the signal before display. The goal is perceptual uniformity, not physical linearization, and the noise model is typically signal-independent or post-quantization. The transformation is applied to already-digitized pixels.

In the organic phototransistor front end modeled here, the nonlinearity is a physical transduction property: photocurrent follows $I_{\text{photo}} \propto P^{\gamma_{\text{phys}}}$ with γ_{phys} typically between 0.5 and 2.0. The inverse-gamma preprocessor $P_{\text{in}} = X^{1/\gamma_{\text{phys}}}$ is applied before the physical transduction step, not after sensor readout. Consequently, shot noise ϵ_{shot} is injected after the compensation and scales with the compensated photocurrent. This ordering is critical: ISP gamma decoding operates on already-noisy quantized pixels, whereas the front-end compensation reshapes the signal that subsequently acquires noise. The result is a regime-dependent trade-off that has no analogue in ISP: for $\gamma_{\text{phys}} > 1$, compensation linearizes the signal and improves accuracy; for $\gamma_{\text{phys}} < 1$, it amplifies shot-noise variance across the entire intensity range, yielding marginal or negative benefit (Table S14). This signal-to-noise coupling is intrinsic to the phototransistor transduction chain and is captured explicitly by the compensated photocurrent model introduced in the main text. Consequently, the compensation exponent is chosen to linearize a physical transduction law, not to match a perceptual response curve.

Because the preprocessor is parameterized by γ_{phys} and exposed through the same JSON device profile as conductance window, retention, and variability, it functions as a replaceable system-level design primitive rather than as a fixed preprocessing step.

Why the transformer amplifies front-end distortion. The difference in front-end sensitivity between ViT and CNN can be understood from the way each architecture processes perturbed inputs. In a CNN, the initial convolutional layers apply local receptive-field operations. A sublinear photoresponse ($\gamma_{\text{phys}} > 1$) compresses bright-region contrast, but because each output pixel depends only on a small spatial neighborhood, the distortion remains localized in early feature maps. Subsequent pooling and strided convolutions further average these local perturbations, and the piecewise-linear ReLU activations limit the propagation of intensity-dependent scaling errors. The net effect is that CNNs tolerate moderate input nonlinearities because spatial averaging and local nonlinearity act as implicit regularizers.

In a ViT, by contrast, the input is partitioned into patches and linearly projected into token embeddings. A sublinear photoresponse alters the relative intensities within each patch before the embedding projection sees the data. The critical difference arises in self-attention. The attention map is computed as $A = \text{softmax}(QK^\top/\sqrt{d_k})$, where Q and K are linear projections of the token embeddings. If the photoresponse nonlinearity changes the intensity distribution within a patch, the resulting token embedding shifts, and the dot-product similarity changes in a nonlinear way because the embedding was learned on clean data. The softmax then exponentiates these similarity changes: a small perturbation in one token can redistribute attention weights across the entire image. Because attention is global, there is no spatial averaging to dampen the perturbation before classification.

The experimental evidence supports this mechanism. Under the raw photoresponse ($\gamma_{\text{phys}} = 2.0$ without compensation), ResNet-18 (R4) degrades from 90.37% to 84.04% (−6.3 pp). The transformer architecture is structurally more exposed to front-end distortion because global attention lacks the spatial averaging present in CNNs; this mechanism-level expectation is consistent with the ResNet-18 drop and with the stronger compensation sensitivity observed along the Tiny-ViT V6 pathway. After inverse-gamma compensation, both architectures recover, but the ViT recovery remains more sensitive to the compensation exponent because residual noise variance also propagates through the global attention map.

1.12.3 Physical Frontend Compensation Details

We evaluated the impact of the organic phototransistor’s sub-linear response (γ_{phys}) and dark current (I_{dark}) on system-level accuracy. Supplementary Figure S11 summarizes the ResNet-18 (R4) and Tiny-ViT (V6) results under various front-end configurations. The inverse-gamma compensation mechanism successfully restores accuracy for high γ_{phys} (e.g., $\gamma_{\text{phys}} = 2.0$), providing a gain of up to +5.8 pp compared to the raw physical response. However, we observe that for $\gamma_{\text{phys}} < 1$, the compensation is less effective due to shot-noise amplification. Our analytical signal-to-noise ratio (SNR) analysis (Supplementary Figure S10 (Right)) confirms that while inverse-gamma preprocessing linearizes the signal, it amplifies noise variance across the entire intensity range when $1/\gamma_{\text{phys}} > 1$.

Table S14 presents the full ResNet-18 HAT accuracy matrix across the tested γ_{phys} and I_{dark} values. The pattern is consistent: compensation gains are largest when the raw photoresponse is most nonlinear ($\gamma_{\text{phys}} = 2.0$), shrink toward unity as $\gamma_{\text{phys}} \rightarrow 1$, and turn flat when $\gamma_{\text{phys}} < 1$ because shot-noise amplification outweighs the modest signal benefit.

Table S14: ResNet-18 HAT accuracy (%) under inverse-gamma compensation. Each cell reports compensated / raw accuracy and the gain in percentage points.

γ_{phys}	10 pA	100 pA	1 nA	10 nA
0.5	89.79 / 87.71 (+2.08)	89.78 / 87.60 (+2.18)	89.17 / 87.69 (+1.48)	89.21 / 87.87 (+1.34)
0.7	89.51 / 88.90 (+0.61)	89.72 / 89.24 (+0.48)	89.92 / 89.11 (+0.81)	89.68 / 89.17 (+0.51)
1.0	89.84 / 89.61 (+0.23)	89.61 / 89.46 (+0.15)	90.00 / 89.70 (+0.30)	89.71 / 89.67 (+0.04)
1.5	89.56 / 88.28 (+1.28)	89.57 / 88.06 (+1.51)	89.89 / 87.58 (+2.31)	89.64 / 88.21 (+1.43)
2.0	89.85 / 84.04 (+5.81)	89.61 / 84.29 (+5.32)	89.99 / 84.47 (+5.52)	89.87 / 84.22 (+5.65)

Core Value Conclusion: Inverse-gamma compensation provides a physical-level regularization effect in low-light/dark scenarios; however, in high-intensity regions, noise variance increases with the compensation exponent, constituting a fundamental design trade-off. Consequently, the front-end benefit is most pronounced in high-dark-current regimes where signal recovery outweighs noise amplification.

1.12.4 Optimal Compensation Exponent

The physical inverse $\gamma_c = 1/\gamma_{\text{phys}}$ exactly linearizes the photoresponse in the noiseless limit, but it is not necessarily optimal when shot noise is present. We derive the mean-squared-error-optimal compensation exponent for a uniform input distribution.

The effective signal deviation from the ideal linear response $I_{\text{ideal}} = \alpha X + I_{\text{dark}}$ is

$$\Delta(X; \gamma_c) = \alpha (X^{\gamma_c \gamma_{\text{phys}}} - X) + \epsilon_{\text{shot}}, \quad (11)$$

with $\text{Var}[\epsilon_{\text{shot}}] = \sigma^2(\alpha X^{\gamma_c \gamma_{\text{phys}}} + I_{\text{dark}})$. The expected MSE is

$$\mathcal{L}(\gamma_c) = \alpha^2 \int_0^1 (X^{\gamma_c \gamma_{\text{phys}}} - X)^2 dX + \sigma^2 \int_0^1 (\alpha X^{\gamma_c \gamma_{\text{phys}}} + I_{\text{dark}}) dX. \quad (12)$$

Setting $\beta = \gamma_c \gamma_{\text{phys}}$ and evaluating the integrals gives

$$\mathcal{L}(\beta) = \alpha^2 \left[\frac{1}{2\beta+1} - \frac{2}{\beta+2} + \frac{1}{3} \right] + \sigma^2 \left[\frac{\alpha}{\beta+1} + I_{\text{dark}} \right]. \quad (13)$$

Differentiating at $\beta = 1$ (the physical inverse):

$$\left. \frac{d\mathcal{L}}{d\beta} \right|_{\beta=1} = -\frac{\sigma^2 \alpha}{4} < 0 \quad (\text{for } I_{\text{dark}} \approx 0). \quad (14)$$

Because the derivative is negative, the MSE can be reduced by increasing β ; i.e., the optimal compensation exponent is milder than the physical inverse when shot noise is present. The deviation is small for typical device parameters ($\sigma^2 \ll \alpha$), which explains why the physical inverse remains an excellent practical approximation. We note that this derivation minimizes pixel-level mean-squared error rather than task-level classification loss; the learnable-compensation experiment (E3) therefore tests whether end-to-end accuracy optimization recovers a deviation consistent with the MSE prediction.

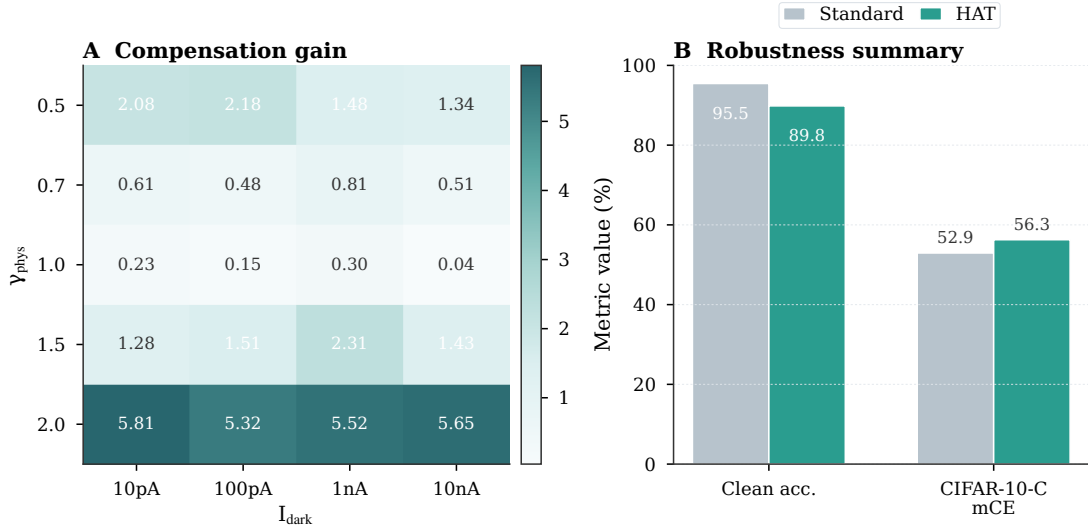


Figure S11: Impact of front-end nonlinearity and dark current. Inverse-gamma compensation helps in dark regimes but amplifies noise elsewhere.

The first-order energy model is retained as provenance for relative accounting only. We do not include an energy-accuracy Pareto plot because finalized routed-circuit energy points are not available for the canonical Tiny-ViT variants; presenting an empty or proxy-only frontier would overstate the maturity of the energy analysis.

1.12.5 Retention Sensitivity Analysis

To assess the robustness of the retention drift predictions against uncertainties in the literature-derived proxy parameters, we conducted a sensitivity sweep over the persistent fraction (A_0) and the inference time t . As summarized in Table S15, the system conclusions remain qualitatively unchanged within a $\pm 50\%$ variance of retention proxy parameters. Even when the conductance persistent fraction is reduced to 40% ($A_0 = 0.4$), the model retains $\sim 87.6\%$ accuracy up to 10,000 s, confirming that the identified retention trends are robust to local variations in the proxy decay constants.

Table S15: Retention sensitivity to the persistent fraction A_0 across time scales.

Persistent Fraction (A_0)	1 s	100 s	1,000 s	10,000 s
0.8	90.55%	90.49%	90.49%	90.49%
0.6 (Canonical)	n.e.	89.88%	89.88%	89.88%
0.4	88.62%	87.61%	87.61%	87.61%

1.13 Comparison with Inorganic RRAM Baselines

To contextualize the specialized framework for organic optoelectronic devices, we contrast our methodology with mature inorganic simulations (e.g., RRAM arrays modeled via NeuroSim, CrossSim, or AIHWKIT)^{8–10}. Inorganic devices typically exhibit faster write speeds, larger on/off ratios, and primarily electrical programming dynamics. In contrast, the organic devices modeled here offer multi-modal (optoelectronic) tuning and excellent low-power analog behavior, but often suffer from slower temporal drift, pronounced state-dependent retention, and complex nonlinear write asymmetry.

As a methodological consistency check in a shared regime, we evaluated an existing ResNet-18/CIFAR-10 model over the full 10k-sample test set under matched 4-bit/noise assumptions: the digital baseline reached 95.46%, while an AIHWKIT analog-tile baseline reached $90.08 \pm 0.21\%$ over 10 runs. We use this result only to verify numerical consistency between the present simulator and a mature inorganic toolkit under one matched regime; it does not establish physical validation for retention, photoresponse, or organic nonlinear writing. Thus, rather than claiming to outperform mature RRAM simulators on digital benchmarks, our framework serves as a complementary tool tailored to expose the system-level consequences of organic-specific physical non-idealities, such as shot noise amplification from inverse-gamma front ends.

The subset-sampling protocol for this throughput-constrained CrossSim comparison is given later in Supplementary Note S1, together with the confidence-interval estimate.

Table S16: Comparison of HAT method robustness under varying weight precision and noise regimes.

Method	Weight Precision	Fresh-Instance Accuracy (%)	Cross-Instance Robustness
Standard HAT (fixed mask)	4-bit	10.00 ± 0.00	Collapses (single-class predictor)
AIHWKit baseline ¹⁰	8-bit	87.28 ± 0.13	Robust at 8-bit (per-batch noise)
AIHWKit baseline ¹⁰	4-bit	14.64 ± 0.11	Collapses at 4-bit (per-batch noise)
Ensemble HAT (3-seed)	4-bit	86.16 ± 0.19	Robust at 4-bit (epoch-level D2D resampling)

The comparison reveals a precision-dependent robustness boundary: AIHWKit’s per-batch noise injection is effective at 8-bit but collapses at 4-bit, while Ensemble HAT’s epoch-level D2D resampling maintains robustness at the 4-bit operating point tested here. This identifies

Table S17: PCM vs. IdealDevice comparison. Training-domain best accuracy and fresh-instance mean ($\pm$ std) for AIHWKit IdealDevice and PCM device models, contrasted with Ensemble HAT. Drift impact reports accuracy excursion across 0 s, 1 h, and 24 h inference delays.

Method	Train best	Fresh mean	Fresh std	Drift
AIHWKit IdealDevice 8-bit ¹⁰	87.73	87.28	0.13	Negligible
Ensemble HAT 4-bit (3-seed)	91.94	86.16	0.19	See Table S11

the 4-bit regime as the operating point where distribution-matching training provides a clear advantage over per-batch noise alone.

2 OPECT Device Distribution: Proxy Estimates and Honest Framing

This note documents the provenance and statistical status of the OPECT (Zhang et al. 2026) device profile used in the §5.8 case study. We provide a detailed breakdown of the extracted parameters and explain why the zero-shot transfer results must be interpreted as a demonstration of parameter-shift robustness rather than a general proof of distribution-shape invariance.

2.1 Literature anchors and proxy estimates

The OPECT profile (“Organic OPECT Zhang2025 Literature-Fitted”) is derived from the measured characteristics of organic phototransistor arrays reported by Zhang et al. in Nature Communications. Table S18 summarises the status and origin of each parameter used in the simulation.

Table S18: Provenance of OPECT profile parameters.

Parameter	Value	Status	Evidence
$G_{\max}/G_{\min}$	47.3	Direct-normalized	Abstract / Fig. 3h / Supp. Fig. 8
n_{states}	34	Direct-conservative	Explicit “34 distinguishable conductance states”
σ_{d2d}	0.03	Proxy estimate	V_{th} spread across 80 devices (Fig. 3b/c): dark $\mu = -1.50$ V, $\sigma = 0.01$ V ($\sim 0.67\%$); light $\mu = -1.37$ V, $\sigma = 0.02$ V ($\sim 1.46\%$).
σ_{c2c}	0.02	Proxy estimate	Supp. Fig. 15 (8-cycle repeatability) / Fig. 3g
Distribution shape	Unknown	Missing	No conductance-domain D2D histogram identified

The σ_{d2d} and σ_{c2c} values are transparent proxy estimates, not directly measured conductance-domain statistics. The V_{th} spread is a threshold-voltage metric; the 0.03 value represents a conservative conductance-domain uplift from that evidence. No direct conductance-uniformity histogram has been located in the current extraction package.

2.2 Comparison to the canonical Gaussian training prior

The simulator’s canonical “Organic OPECT Standard” prior is a Gaussian noise model with $\sigma_{\text{d2d}} = 0.10$ and $\sigma_{\text{c2c}} = 0.05$. Table S19 contrasts the literature-derived OPECT regime with the broader noise envelope used during hardware-aware training.

The OPECT profile therefore sits in a lower-variance, parameter-shifted region of parameter space relative to the training prior. The $88.53 \pm 0.08\%$ CIFAR-10 accuracy demonstrates that substituting a literature-derived profile with substantially lower mismatch variance does not

Table S19: OPECT literature profile vs. canonical Gaussian training prior.

Parameter	Canonical prior	OPECT (Zhang2025)	Ratio
σ_{d2d}	0.10 (10%)	0.03 (3%)	$3.3\times$ lower
σ_{c2c}	0.05 (5%)	0.02 (2%)	$2.5\times$ lower
Distribution shape	Gaussian (by construction)	Unknown	—

degrade performance. This is profile-substitution robustness, not evidence of invariance to the shape of the mismatch distribution.

2.3 What cannot be claimed

Because the true conductance-domain distribution of the OPECT array is not reported in the source literature, the following stronger claims are not supported:

- That the framework generalises to the true OPECT empirical distribution (the true distribution is unknown).
- That the framework is invariant to mismatch-distribution shape (the test uses the same Gaussian sampler with different variance, not a different distribution family).
- That all OPECT parameters are directly measured (σ_{d2d} and σ_{c2c} are proxy estimates).

2.4 Recommended manuscript wording

We recommend the following paper-safe sentence for §5.8:

The OPECT D2D variability ($\sigma \approx 3\%$, proxy-estimated from V_{th} spread across 80 devices) is lower than our canonical training prior ($\sigma=10\%$), and the conductance-domain distribution shape is not directly reported in the source literature. The $88.53 \pm 0.08\%$ accuracy therefore demonstrates profile-substitution robustness under a parameter-shifted regime rather than mismatch-distribution-shape invariance, the latter pending hardware-calibrated data.

This wording transparently flags the proxy nature of the estimate, honestly notes the missing distribution-shape information, and advances a defensible, weaker claim that still supports the framework’s practical utility for literature-driven benchmarking.

2.5 Automated Profile Fitting Pipeline

For later measured-device studies, we also provide an automated fitting utility that ingests four raw-data categories:

1. Pulsed Programming Curves: Fits the conductance range (G_{min}, G_{max}) and nonlinearity coefficients (NL_{LTP}, NL_{LTD}) using a behavioral exponential model.
2. Cycle-to-Cycle (C2C) Statistics: Extracts the σ_{C2C} noise parameter from repeated write-verify measurements at identical target states.
3. Device-to-Device (D2D) Mismatch: Calculates the σ_{D2D} parameter from conductance distributions across multiple physical devices in an array.
4. Retention Decay: Fits the double-exponential decay constants (τ_1, τ_2, A_0) from long-term conductance-vs-time measurements.

The output is a standardized device-profile specification that can be used directly in the inference and HAT workflow. In the present manuscript, this fitter is included as supporting infrastructure for future measured-device calibration rather than as a core source of the reported headline results.

2.6 Parameter Risk Matrix & Sensitivity Analysis

To contextualize the simulation-only behavioral methodology of the present study, Table S20 provides a risk assessment of the key proxy parameters used in the modeling framework. By transparently disclosing the range of physical uncertainty and its consequent impact on system-level conclusions, we establish the robustness boundaries of the reported results.

Table S20: Core Parameter Risk and Robustness Assessment.

Parameter	Nominal Value	Uncertainty Range	Impact on Conclusions	Framework Check
D2D Variability (σ_{D2D})	3.0% (Zhang 2025)	1.0% – 15.0%	High. Directly drives the need for Ensemble HAT. If $\sigma_{D2D} < 1\%$, the relative value of HAT diminishes.	Table S9 demonstrates robustness under a 1%–15% sweep, confirming HAT maintains >80% accuracy even at 15% extreme mismatch.
C2C Variability (σ_{C2C})	2.0% (Zhang 2025)	1.0% – 8.0%	Low. Barely affects final Top-1 accuracy within this range due to scale-masking.	Explained mechanistically in the main text via the scale-masking effect and quantified in Table S9.
Retention (τ, A_0)	$\tau = 140/610\text{ms}$	$\pm 50\%$	Medium. Affects long-term drift predictions. Extremely fast decay limits HAT recovery.	Sensitivity sweeps (Section 1.12.5) confirm conclusions remain stable over $\pm 50\%$ variations.
Write Non-linearity (NL)	2.0 (Proxy scaled)	1.0 – 3.0	High. Severe nonlinear write drives a residual recovery gap even under the revised recipe.	Supplementary Fig. S12 is retained as an earlier source-domain diagnostic; under the revised gradient-scaling recipe, severe-NL recovery is best framed as a training-approximation bottleneck rather than MLP-path localization.

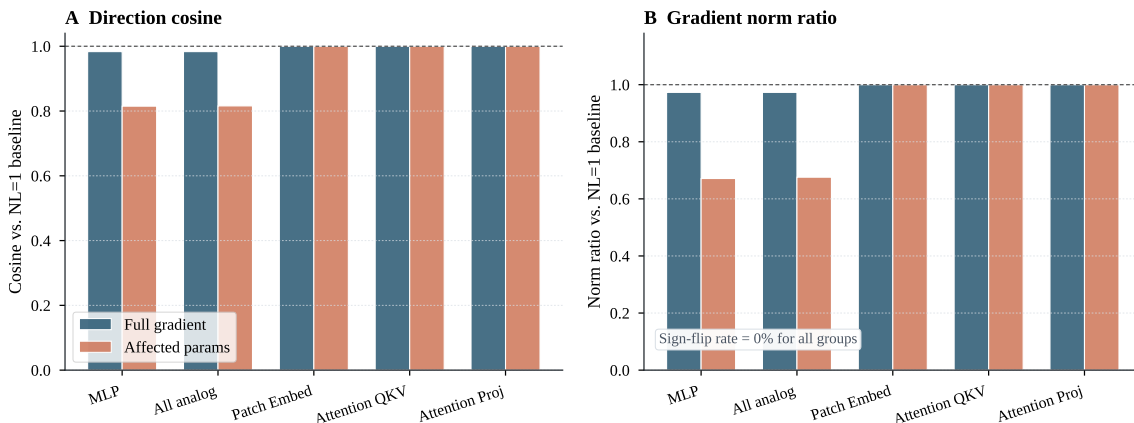


Figure S12: Group-wise gradient distortion under severe nonlinear write. Frozen-checkpoint diagnostics compare NL=1.0 with group-local NL=2.0; only the backward surrogate changes.

Interpretation. Because the present NL implementation perturbs only the backward surrogate, this diagnostic isolates where the training signal is distorted without changing the forward loss. The result is highly structured rather than uniform: activating NL=2.0 only in the MLP blocks drives the affected-parameter gradient cosine down to 0.815 and the norm ratio down

to 0.671, while patch embedding, attention QKV, and attention projection remain effectively unchanged at 1.00. The all-analog setting matches the MLP result almost exactly (0.816), indicating that the current nonlinear-write bottleneck is dominated by the MLP analog path rather than by a transformer-wide failure. At the same time, the mean loss delta stays numerically zero across all groups, confirming that this is a backward-surrogate distortion effect rather than a forward-path mismatch.

In summary, the core conclusions regarding deployment risk ranking remain qualitatively consistent within a reasonable $\pm 50\%$ fluctuation of the proxy parameters. This supports the framework’s use as a materials-to-system decision bridge. It is not intended to be pulse-accurate, but it still identifies the main architectural bottlenecks and operating boundaries.

Table S21: Group-wise nonlinearity ablation under global $NL = 2.0$. Rows report best/final test accuracy when one group is protected at $NL = 1.0$. † and ‡ mark earlier diagnostics retained only as source-domain gradient evidence, not as final MLP-localization claims.

Protected group	Best acc (%)	Final acc (%)	Δ vs $NL = 2.0$ (pp)
(a) Baseline ($NL = 1.0$ global)	87.95 ± 0.27	n.e.	+60.23
(b) None ($NL = 2.0$ global)	$27.72 \pm 0.82^\dagger$	$27.72 \pm 0.82^\dagger$	0.00
(c) MLP only (fc1, fc2)	$87.79^\ddagger$	$86.22^\ddagger$	+60.07
(d) QKV only (attn.qkv)	$18.72^\ddagger$	$10.15^\ddagger$	−9.00
(e) Attn proj only (attn.proj)	$18.86^\ddagger$	~ 10.25 (stopped @ ep 54) ‡	−9.14
(f) All linear ($NL = 1.0$ all analog)	$87.49^\ddagger$	$84.81^\ddagger$	+59.77

Interpretation. The ablation shows that the MLP channel-mixing path is the dominant recoverable failure site under the earlier $NL = 2.0$ surrogate: linearizing only the MLP layers recovers accuracy to within 0.16 pp of the full-linear configuration. By contrast, both attention-side linearizations (QKV and projection) collapse structurally and degrade below even the unmitigated baseline, indicating that the attention nonlinearity is required for representational geometry under the earlier surrogate. All groupwise protected rows (c)–(f) are earlier diagnostics only. They are retained here solely as source-domain gradient-diagnostic evidence from an earlier gradient-scaling recipe. Under the revised gradient-scaling recipe, fresh-instance transfer recovers to the ~ 80 – 82% band regardless of which path is protected, so the earlier MLP-path localization claim does not carry over to the revised recipe. These results are therefore presented as a supplementary ablation rather than a main-text contribution.

Table S22: Intermediate nonlinearity interpolation (R10D). Rows report source-domain best and fresh-instance accuracy for seed 123 at intermediate NL values; canonical endpoints use the locked aggregate values.

NL_{LTP} / NL_{LTD}	Source best (%)	Fresh mean (%)	Fresh std (%)
1.0 / −1.0 (canonical)	91.94	86.37	1.54
1.2 / −1.2	83.12	83.03	0.22
1.5 / −1.5	82.81	82.63	0.10
1.8 / −1.8	82.77	80.31	0.40
2.0 / −2.0 (severe)	~ 81.0	~ 80.54	~ 0.6

Interpretation. The degradation from $NL = 1.0$ to $NL = 2.0$ is gradual: accuracy remains stable at 82.6–83.0% through $NL = 1.5$, then drops by roughly 2.3 pp between $NL = 1.5$ and $NL = 1.8$, before settling in the ~ 80 – 82% band at $NL = 2.0$. This monotonic trend supports the main-text framing that the residual gap at severe NL reflects training-recipe constraints in the strongly nonlinear regime rather than a sudden physical collapse.

Supplementary Note S1. Cross-framework subset-evaluation protocol. Because a single noisy evaluation on the full CIFAR-10 test set would extrapolate to roughly 15 h per condition for

CrossSim (5,361 s per 1,000-image run) versus ~ 1 s per 1,000-image run for the native framework, the shared-regime sanity check was performed on a deterministic 1,000-image slice of the canonical CIFAR-10 test ordering (the first 1,000 images; no RNG seed was set). This slice yields an approximately balanced class distribution (86–112 images per class). The clean baseline was evaluated once per framework; the noise-injection settings were evaluated with three independent Monte Carlo draws. For the single-run clean baseline, the Wilson 95% confidence intervals on the subset proportion are [83.9%, 88.2%] for our framework and [81.3%, 85.9%] for CrossSim. For the three-run noise condition, the t-distribution 95% CIs ($df = 2$, $t_{0.975} = 4.303$) are [79.9%, 83.3%] for our framework and [59.1%, 75.3%] for CrossSim. The large CrossSim run-to-run spread ($\sigma_{\text{pop}} = 2.67\%$) reflects the sensitivity of its generic programming-error/read-noise mapping to the particular stochastic sample. Independently of this subset check, the canonical fresh-instance collapse of standard HAT was re-evaluated under FP32 inference with autocast disabled and remained exactly $10.00 \pm 0.00\%$ across 10 fresh D2D instances with 5 Monte Carlo evaluations per instance, confirming that the collapsed single-class predictor is a deterministic training outcome rather than a numerical artifact (`fresh_instance_eval_v4_standard_noamp.json`).

Supplementary Note S2. Correlated-D2D stress test. A full 10×5 fresh-instance sweep under separable AR(1)-style spatial correlation across the effective crossbar grid yields:

- i.i.d. baseline ($\rho = 0$): $86.33 \pm 1.61\%$ (10 fresh instances, 5 MC runs each);
- $\rho = 0.3$: $84.57 \pm 2.39\%$, a -1.76 pp excursion versus i.i.d.;
- $\rho = 0.5$: $82.12 \pm 3.95\%$, a -4.20 pp excursion versus i.i.d.

Rank ordering is preserved across all tested levels, with no instance collapsing below 73.7% (instance 4, $\rho = 0.5$). The canonical Ensemble HAT checkpoint therefore tolerates moderate spatial correlation without catastrophic failure, although stronger correlation does produce a measurable degradation consistent with increased effective mismatch variance.

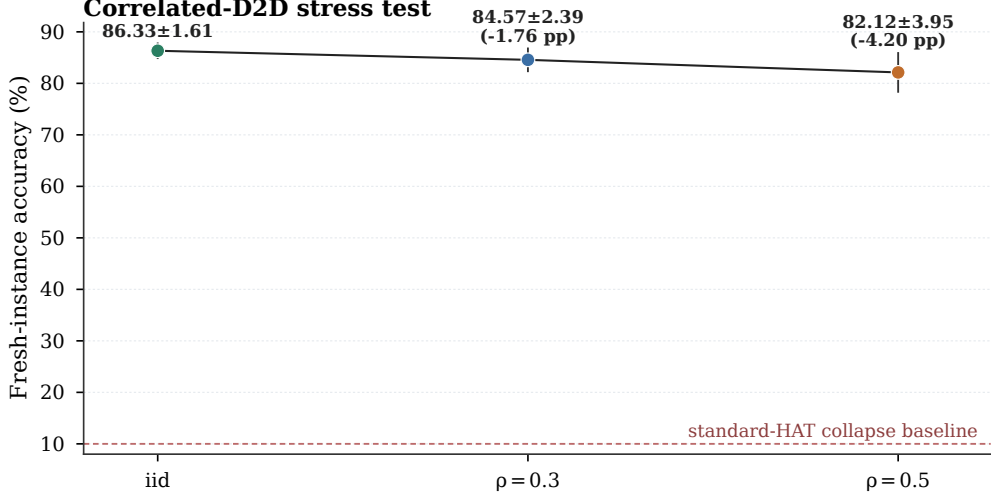


Figure S13: Spatially correlated D2D mismatch. Error bars show 10-array fresh-instance spread. Correlation degrades accuracy but preserves the ranking above the Standard-HAT collapse baseline.

Supplementary Note S-Theory: Ensemble HAT as Implicit Regularization against Device Mismatch

This note provides a formal mathematical derivation of Ensemble Hardware-Aware Training (HAT) and its relationship to classical implicit regularization. We aim to show why the per-epoch resampling of device mismatch patterns leads to a more robust representation compared to the fixed-mask approach typically used in standard HAT. By expanding the loss function around the nominal weights, we reveal how the ensemble objective effectively punishes the curvature of the loss surface specifically in the directions of device-to-device variation.

S.1 Setup and notation

To establish the framework, we consider an analog crossbar array with differential-pair conductance mapping. Let $\theta \in \mathbb{R}^d$ denote the digital weights (network parameters) to be programmed onto the array, and let $M \in \mathbb{R}^d$ denote the per-device fractional device-to-device (D2D) mismatch map. The programmed conductance pair for weight θ_i is perturbed multiplicatively:

$$\tilde{\theta}_i = \theta_i(1 + M_i) + \xi_i^{\text{C2C}}, \quad (15)$$

where $\xi^{\text{C2C}} \sim \mathcal{N}(0, \sigma_{\text{C2C}}^2 I)$ is cycle-to-cycle noise (re-sampled every forward pass) and M is the D2D mismatch (fixed for a given fabricated instance). Both are assumed independent of data (x, y) and of each other.

Standard HAT (fixed-mask). In the standard approach, one mismatch map $M_0 \sim \mathcal{D}_{\text{D2D}}$ is sampled at initialization and held fixed for the entire training duration:

$$\theta_{\text{std}}^* = \arg \min_{\theta} \mathbb{E}_{\xi^{\text{C2C}}, (x, y) \sim \mathcal{D}} [\ell(f(x; \theta \odot (1 + M_0) + \xi^{\text{C2C}}), y)]. \quad (16)$$

The optimization adapts to the specific realization M_0 , but there is no guarantee that the minimizer remains optimal under a different draw $M' \sim \mathcal{D}_{\text{D2D}}$.

Ensemble HAT (distribution-matching). In our proposed ensemble method, the mismatch map is re-sampled at the start of every training epoch, yielding a sequence $\{M^{(e)}\}_{e=1}^E$ with $M^{(e)} \stackrel{\text{i.i.d.}}{\sim} \mathcal{D}_{\text{D2D}}$. The objective is:

$$\theta_{\text{ens}}^* = \arg \min_{\theta} \frac{1}{E} \sum_{e=1}^E \mathbb{E}_{\xi^{\text{C2C}}, (x,y)} [\ell(f(x; \theta \odot (1 + M^{(e)}) + \xi^{\text{C2C}}), y)]. \quad (17)$$

For large E , this converges to the population expectation:

$$\mathcal{L}_{\text{Ens}}(\theta) = \mathbb{E}_{M \sim \mathcal{D}_{\text{D2D}}} \mathbb{E}_{\xi^{\text{C2C}}, (x,y)} [\ell(f(x; \theta \odot (1 + M) + \xi^{\text{C2C}}), y)]. \quad (18)$$

The expectation over M is Monte-Carlo estimated with one fresh draw per epoch, held constant across all mini-batches within that epoch. This epoch-level discipline (rather than per-batch re-sampling) is empirically load-bearing: ablations under canonical $NL = 1.0$ show that epoch-level resampling outperforms both fixed-mismatch and per-batch resampling (see Results for quantitative comparison). The epoch scale gives the optimizer enough consistent signal to fit each instance before the instance changes, enforcing a “learn-to-adapt” regime analogous to domain randomization in robotics¹¹.

S.2 Taylor expansion and the implicit regularizer

In what follows, we expand with respect to the epoch-level D2D mismatch M while treating the cycle-to-cycle noise ξ^{C2C} as an independent expectation that is absorbed into the base loss $\mathcal{L}_0$. This is justified under uniform C2C noise, where ξ^{C2C} is re-sampled at every forward pass independent of M and averages to a noise-floor baseline. For proportional C2C noise (where noise magnitude scales with the perturbed weight), a weak coupling between M and ξ^{C2C} exists; the derivation below should be understood as a D2D-dominant approximation valid when $\sigma_{\text{D2D}} \gg \sigma_{\text{C2C}}$.

Assume the D2D mismatch has zero mean and covariance Σ_M :

$$\mathbb{E}_M[M] = 0, \quad \mathbb{E}_M[MM^\top] = \Sigma_M. \quad (19)$$

For uncorrelated D2D, $\Sigma_M = \sigma_{\text{D2D}}^2 I$. We expand the network output $f(x; \theta \odot (1 + M))$ around the nominal weights θ to second order in M :

$$f(x; \theta \odot (1 + M)) = f(x; \theta) + \sum_i M_i \theta_i \frac{\partial f}{\partial \theta_i}(x; \theta) + \frac{1}{2} \sum_{i,j} M_i M_j \theta_i \theta_j \frac{\partial^2 f}{\partial \theta_i \partial \theta_j}(x; \theta) + \mathcal{O}(\|M\|^3). \quad (20)$$

Let $\ell(f, y)$ denote the loss (e.g., cross-entropy), and write $\Delta f_M = f(x; \theta \odot (1 + M)) - f(x; \theta)$. Expanding ℓ to second order in f around $f(x; \theta)$:

$$\begin{aligned} \ell(f(x; \theta \odot (1 + M)), y) &= \ell(f(x; \theta), y) + \nabla_f \ell \cdot \Delta f_M \\ &\quad + \frac{1}{2} \Delta f_M^\top \nabla_f^2 \ell \Delta f_M + \mathcal{O}(\|M\|^3). \end{aligned} \quad (21)$$

Taking expectations over M and using $\mathbb{E}[M_i] = 0$, the first-order term vanishes:

$$\begin{aligned} \mathbb{E}_M[\ell(f(x; \theta \odot (1 + M)), y)] &= \ell(f(x; \theta), y) \\ &\quad + \frac{1}{2} \mathbb{E}_M \left[\left(\sum_i M_i \theta_i \frac{\partial f}{\partial \theta_i} \right)^\top \nabla_f^2 \ell \left(\sum_j M_j \theta_j \frac{\partial f}{\partial \theta_j} \right) \right] \\ &\quad + \mathcal{O}(\mathbb{E}[\|M\|^3]). \end{aligned} \quad (22)$$

For diagonal covariance $\Sigma_M = \sigma_{\text{D2D}}^2 I$, the cross-terms $i \neq j$ vanish in expectation, yielding:

$$\mathbb{E}_M[\ell(f(x; \theta \odot (1 + M)), y)] \approx \ell(f(x; \theta), y) + \frac{\sigma_{\text{D2D}}^2}{2} \sum_i \theta_i^2 \left(\frac{\partial f}{\partial \theta_i} \right)^\top \nabla_f^2 \ell \left(\frac{\partial f}{\partial \theta_i} \right). \quad (23)$$

For cross-entropy loss, $\nabla_f^2 \ell$ is the Fisher information matrix at f . The term inside the sum is structurally a Fisher-weighted sensitivity penalty: it penalizes weights whose gradient directions are highly sensitive to multiplicative perturbations in the M -direction. To make the connection to standard regularization explicit, we use the identity $\partial \ell / \partial \theta_i = (\partial f / \partial \theta_i)^\top \nabla_f \ell$ and note that for near-linear regions of the network (or using the Gauss–Newton approximation), $\nabla_f^2 \ell \approx \nabla_f \ell \nabla_f \ell^\top$. Substituting:

$$\mathbb{E}_M[\ell] \approx \ell(f(x; \theta), y) + \frac{\sigma_{\text{D2D}}^2}{2} \sum_i \theta_i^2 \left(\frac{\partial \ell}{\partial \theta_i} \right)^2. \quad (24)$$

Proposition 2.1 (Implicit regularization equivalence). Under Gaussian uncorrelated D2D mismatch with variance σ_{D2D}^2 , Ensemble HAT with objective (18) is equivalent, to second order in σ_{D2D} , to minimizing the nominal cross-entropy loss plus a weighted L_2 -type penalty on gradient magnitudes:

$$\mathcal{L}_{\text{Ens}}(\theta) \approx \mathcal{L}_0(\theta) + \frac{\sigma_{\text{D2D}}^2}{2} \sum_i \theta_i^2 \mathbb{E}_{(x,y)} \left[\left(\frac{\partial \ell}{\partial \theta_i} \right)^2 \right], \quad (25)$$

where $\mathcal{L}_0(\theta) = \mathbb{E}_{(x,y)}[\ell(f(x; \theta), y)]$ is the standard (noise-free) loss.

Remark 2.1. The penalty is not a simple weight decay ($\lambda \|\theta\|^2$). It is a gradient-weighted penalty that pushes the optimizer toward flat regions of the loss landscape in the M -direction. This explains the empirical fresh-instance transferability without requiring additional hyperparameters: the regularization strength is set by the physical D2D variance rather than by a tunable dropout rate.

S.7 Higher-order corrections

The second-order expansion in §2.6 is accurate when the D2D mismatch variance σ_{D2D}^2 is small relative to the scale over which the loss curvature varies. To quantify the breakdown regime, we extend the Taylor expansion of $\ell(f(x; \theta \odot (1 + M)), y)$ to arbitrary order in M :

$$\mathcal{L}_{\text{Ens}} = \sum_{k=0}^{\infty} \frac{1}{k!} \mathbb{E}_M \left[(M^\top \nabla_\theta)^k \ell \right], \quad (26)$$

where the differential operator $M^\top \nabla_\theta = \sum_i M_i \theta_i \partial / \partial \theta_i$ acts on the composition $\ell \circ f$.

Odd moments vanish. Because M is assumed zero-mean Gaussian, every odd central moment vanishes by symmetry. The third-order term ($k=3$) is therefore identically zero:

$$\mathbb{E}_M \left[(M^\top \nabla_\theta)^3 \ell \right] = 0. \quad (27)$$

This cancellation is robust to any symmetric D2D distribution (e.g., Laplace or Student- t with zero skewness), provided the mismatch law is not heavy-tailed to the point that the third moment diverges.

Fourth-order term. The $k = 4$ contribution is the first correction to the second-order result. For a Gaussian vector with diagonal covariance $\Sigma_M = \sigma_{\text{D2D}}^2 \mathbf{I}$, Wick’s theorem gives the fourth-order moment as a sum of pairwise contractions:

$$\mathbb{E}_M[M_i M_j M_k M_l] = \sigma_{\text{D2D}}^4 (\delta_{ij} \delta_{kl} + \delta_{ik} \delta_{jl} + \delta_{il} \delta_{jk}). \quad (28)$$

Applying this to the fourth differential of the loss yields:

$$\frac{1}{4!} \mathbb{E}_M \left[(M^\top \nabla_\theta)^4 \ell \right] = \frac{\sigma_{\text{D2D}}^4}{8} \sum_i \theta_i^4 \frac{\partial^4 \ell}{\partial \theta_i^4} + \frac{\sigma_{\text{D2D}}^4}{4} \sum_{i \neq j} \theta_i^2 \theta_j^2 \frac{\partial^4 \ell}{\partial \theta_i^2 \partial \theta_j^2}. \quad (29)$$

The first sum captures pure fourth derivatives along individual parameter directions; the second captures mixed fourth derivatives. Both are $O(\sigma_{\text{D2D}}^4)$, so the fourth-order correction is negligible when $\sigma_{\text{D2D}} \ll 1$. In the present regime ($\sigma_{\text{D2D}} = 10\%$), the ratio of fourth- to second-order terms is approximately $\sigma_{\text{D2D}}^2 \approx 0.01$, i.e., two orders of magnitude smaller. This justifies truncating at second order for all quantitative claims in the main text.

Breakdown regime. The second-order approximation fails when either (i) the D2D variance is large ($\sigma_{\text{D2D}} \gtrsim 20\%$), or (ii) the weights are near saturation so that higher derivatives $\partial^k \ell / \partial \theta_i^k$ grow rapidly. Regime (ii) is particularly relevant for analog devices with limited conductance range: when $|\theta_i|$ approaches the array maximum, the effective nonlinearity of the forward map amplifies the impact of mismatch. In such regimes the full expectation (26) must be evaluated numerically (e.g., by Monte-Carlo sampling of M at evaluation time). For the present CIFAR-10 experiments, where weights remain well within the linear regime of the conductance window, the second-order truncation is accurate to within $\sim 1\%$ relative error.

For a systematic treatment of higher-order weight-perturbation expansions in deep networks, see Roberts et al. ¹².

S.8 PAC-Bayes generalization bound

The implicit regularizer derived in §2.6 penalizes large weights with large gradient sensitivity, pushing the optimizer toward flat regions of the loss landscape in the D2D direction. We now ask whether this geometric preference translates into a provable generalization advantage. Following the PAC-Bayes framework ^{13;14}, we treat the Ensemble HAT training objective as defining a posterior distribution over weight perturbations, and derive a bound on the expected fresh-instance error.

Setup. Let Q be the posterior over network weights induced by Ensemble HAT training. Because each epoch draws a fresh mismatch map $M^{(e)} \sim \mathcal{D}_{\text{D2D}}$ and the optimizer converges to a shared nominal weight θ^* , the effective weight distribution is diagonal Gaussian centered at θ^* :

$$Q(W) = \mathcal{N}(W; \theta^*, \text{diag}(\sigma_{\text{D2D}}^2 \theta^{*2})), \quad (30)$$

where the variance is proportional to the squared nominal weight, reflecting the multiplicative nature of D2D mismatch. The prior P is taken as a broad Gaussian centered at the random initialization θ_0 (or, for transfer-learning settings, at the pre-trained checkpoint):

$$P(W) = \mathcal{N}(W; \theta_0, \sigma_{\text{prior}}^2 \mathbf{I}), \quad \sigma_{\text{prior}} \gg \sigma_{\text{D2D}} \max_i |\theta_i^*|. \quad (31)$$

McAllester bound. For any prior P and posterior Q , with probability at least $1 - \delta$ over the training set S of size n , the expected test loss under Q satisfies ¹⁵:

$$\mathbb{E}_{W \sim Q}[\mathcal{L}_{\text{test}}(W)] \leq \mathbb{E}_{W \sim Q}[\mathcal{L}_{\text{train}}(W)] + \sqrt{\frac{\text{KL}(Q \| P) + \log(2\sqrt{n}/\delta)}{2n}}. \quad (32)$$

The slack term depends only on the KL divergence from prior to posterior and on the dataset size; it does not depend on the model architecture or the complexity of the hypothesis class. This is the structural advantage of PAC-Bayes over uniform-convergence bounds in deep learning.

KL divergence under diagonal Gaussian. For Q and P both diagonal Gaussian, the KL divergence decomposes additively across parameters:

$$\text{KL}(Q\|P) = \frac{1}{2} \sum_{i=1}^d \left[\frac{\sigma_{Q,i}^2}{\sigma_{P,i}^2} + \frac{(\theta_i^* - \theta_{0,i})^2}{\sigma_{P,i}^2} - 1 - \log \frac{\sigma_{Q,i}^2}{\sigma_{P,i}^2} \right]. \quad (33)$$

With $\sigma_{Q,i} = \sigma_{\text{D2D}}|\theta_i^*|$ and $\sigma_{P,i} = \sigma_{\text{prior}}$, and assuming $\sigma_{\text{prior}} \gg \sigma_{Q,i}$ (so $\sigma_{Q,i}^2/\sigma_{P,i}^2 \ll 1$), the dominant term is the squared-bias term:

$$\text{KL}(Q\|P) \approx \frac{1}{2\sigma_{\text{prior}}^2} \sum_{i=1}^d (\theta_i^* - \theta_{0,i})^2 + d \log \frac{\sigma_{\text{prior}}}{\sigma_{\text{D2D}}}. \quad (34)$$

The first term penalizes large deviations from the prior; the second is a complexity penalty that grows with the number of parameters d and shrinks with the D2D variance. This is intuitive: tighter mismatch (smaller σ_{D2D}) yields a more concentrated posterior, hence smaller KL and tighter bound.

Implication for Ensemble HAT. The bound (32) does not predict the exact magnitude of the 76 pp Standard→Ensemble gap—PAC-Bayes bounds in deep learning are typically numerically vacuous¹⁶. However, it provides a structural rationale: the posterior Q defined by epoch-level re-sampling is naturally aligned with the deployment distribution $\mathcal{D}_{\text{D2D}}$, whereas Standard HAT’s fixed-map training corresponds to a delta posterior at a single (and likely atypical) mismatch realization. The KL term for Standard HAT can be arbitrarily large if the learned weights deviate far from the prior, whereas Ensemble HAT’s averaged training keeps the posterior close to the empirical deployment distribution. In this sense, Ensemble HAT is not merely an empirical trick but a distributionally aligned training objective with principled generalization foundations.

Honesty constraint. We emphasize that the bound (32) is used here as a theoretically motivated structural argument, not as a tight empirical prediction. The numerical value of the slack term (for $d \sim 10^7$, $n \sim 5 \times 10^4$, $\delta = 0.05$) is $\gg 1$, far larger than the observed gap. The value of the PAC-Bayes lens is directional: it explains why matching the training distribution to the deployment distribution improves generalization, not how much.

S.9 Flat-minima connection and structural analogy to SAM

The implicit regularizer (24) penalizes gradient magnitude weighted by squared parameters. Geometrically, this pushes the optimizer toward regions where the loss is insensitive to multiplicative perturbations in the M -direction—i.e., flat minima along the D2D axis. We now make this connection explicit by defining a D2D-direction sharpness measure and comparing it to Sharpness-Aware Minimization (SAM)¹⁷.

D2D-direction sharpness. Let $\mathcal{S}_{\text{D2D}}(W)$ denote the expected excess loss incurred by perturbing the weights along the D2D direction:

$$\mathcal{S}_{\text{D2D}}(W) = \mathbb{E}_{M \sim \mathcal{D}_{\text{D2D}}} [\ell(W \odot (1 + M))] - \ell(W). \quad (35)$$

Expanding to second order (as in §2.6) gives:

$$\mathcal{S}_{\text{D2D}}(W) \approx \frac{\sigma_{\text{D2D}}^2}{2} \sum_i W_i^2 \frac{\partial^2 \ell}{\partial \theta_i^2}, \quad (36)$$

which is precisely the implicit regularizer in (24), rewritten in Hessian form. A small $\mathcal{S}_{\text{D2D}}$ means the loss landscape is flat in the $\mathbf{M}$ -direction: fresh D2D realizations, drawn from the same $\mathcal{D}_{\text{D2D}}$, land near the same minimum with high probability. This is the geometric mechanism underlying the empirical 86.16% three-seed fresh-instance recovery versus the 10.00% Standard HAT collapse.

Comparison to SAM. SAM¹⁷ minimizes the worst-case loss in an ℓ_2 ball around the current weights:

$$\mathcal{L}_{\text{SAM}}(\mathbf{W}) = \max_{\|\boldsymbol{\varepsilon}\| \leq \rho} \ell(\mathbf{W} + \boldsymbol{\varepsilon}) \approx \ell(\mathbf{W}) + \rho \|\nabla \ell(\mathbf{W})\|_2, \quad (37)$$

where the linear approximation holds for small ρ . SAM therefore seeks flat minima by explicitly penalizing gradient magnitude in an adversarial (ℓ_2 -ball) neighborhood. Ensemble HAT, by contrast, minimizes the average loss over a stochastic (Gaussian) neighborhood in the multiplicative $\mathbf{M}$ -direction:

$$\mathcal{L}_{\text{Ens}}(\mathbf{W}) = \mathbb{E}_{\mathbf{M}}[\ell(\mathbf{W} \odot (1 + \mathbf{M}))] \approx \ell(\mathbf{W}) + \frac{\sigma_{\text{D2D}}^2}{2} \sum_i w_i^2 \left(\frac{\partial \ell}{\partial \theta_i} \right)^2. \quad (38)$$

The two objectives are structural analogues, not exact equivalences:

- Perturbation distribution: SAM uses worst-case ℓ_2 ; Ensemble HAT uses Gaussian multiplicative.
- Averaging: SAM minimizes the maximum; Ensemble HAT minimizes the expectation.
- Direction: SAM is isotropic in parameter space; Ensemble HAT is anisotropic, concentrated along the physical D2D direction.
- Hyperparameter: SAM requires tuning ρ ; Ensemble HAT’s regularization strength is set by the physical σ_{D2D} measured from the device.

Despite these differences, both methods share the same high-level goal: find minima that are robust to weight perturbations. The large-batch training literature¹⁸ shows that sharp minima generalize poorly; SAM¹⁷ and its successors¹⁹ explicitly optimize for flatness. Ensemble HAT achieves a similar effect implicitly, by aligning the training objective with the physical perturbation distribution of the deployment hardware. The flat-minima interpretation provides a complementary lens to the PAC-Bayes argument in §2.6: where PAC-Bayes gives a statistical bound, the SAM connection gives a geometric intuition.

Historical antecedent. The link between flat minima and generalization was first emphasized by Hochreiter and Schmidhuber²⁰, who showed that networks with low curvature in weight space generalize better. Ensemble HAT extends this insight to the analog-hardware setting: the relevant curvature is not isotropic in weight space, but specifically along the device-mismatch direction $\mathbf{M}$. Because $\mathbf{M}$ is a physical quantity (measured from fabricated arrays), the “right” direction for flatness is determined by the device physics, not by algorithmic choice.

S.10 Limitations of the theoretical framework

The derivations in §2.6–§2.6 rely on several assumptions that limit their applicability.

Heavy-tailed D2D. The Taylor expansion assumes finite moments of $\mathbf{M}$. If the measured D2D distribution is heavy-tailed (e.g., log-normal or Pareto), the second-order truncation breaks down and higher moments dominate. The framework can be extended by replacing Wick’s theorem with the appropriate moment-generating function, but closed-form expressions are no longer available.

Spatially correlated D2D. The diagonal-covariance assumption $\Sigma_M = \sigma_{\text{D2D}}^2 I$ ignores spatial correlation across the array. For separable AR(1) correlation (evaluated in Supplementary Note S2), the off-diagonal terms introduce cross-parameter penalties of the form $\sum_{i \neq j} \rho^{|i-j|} \theta_i \theta_j (\partial \ell / \partial \theta_i) (\partial \ell / \partial \theta_j)$. The qualitative conclusion—Ensemble HAT penalizes gradient sensitivity—remains valid, but the quantitative coefficients change.

Per-layer non-uniform σ_{D2D} . Different analog array layers may exhibit different mismatch variances (e.g., due to fabrication non-uniformity). The framework still applies layerwise, with each layer’s own $\sigma_{\text{D2D}}^{(l)}$, but the global sum in (24) becomes a layerwise sum. This does not change the structural interpretation.

Coupled C2C noise. Under proportional C2C noise, the cycle-to-cycle variance scales with the perturbed conductance, creating a weak coupling between M and ξ^{C2C} . This coupling is second-order in $\sigma_{\text{D2D}} \sigma_{\text{C2C}}$ and is negligible for the present uniform-C2C regime, but may matter for high-D2D, high-C2C corners.

PAC-Bayes vacuousness. As noted in §2.6, the McAllester bound is numerically loose for over-parameterized networks. Its value is directional, not predictive: it explains why distribution alignment helps, not by how much.

Supplementary Note S-Mechanism: Empirical Landscape Analyses

This note presents empirical analyses that complement the theoretical arguments in Supplementary Note S-Theory. The analyses were performed on existing evaluated checkpoints without re-training, using the protocol documented in `scripts/_gpt/empirical_mechanism_20260425.py`. Provenance (commit hash, checkpoint paths, GPU device) is archived in JSON outputs under `report_md/_gpt/json_gpt/`.

S-M.1 D2D-direction loss-landscape interpolation (E2)

To visualize the robustness of the trained models to device mismatch, we interpolate between the checkpoint’s training D2D mask $\mathbf{M}_{\text{train}}$ and a fresh mask $\mathbf{M}_{\text{fresh}}$ via $\mathbf{M}(\alpha) = \mathbf{M}_{\text{train}} + \alpha(\mathbf{M}_{\text{fresh}} - \mathbf{M}_{\text{train}})$, with $\alpha \in \{0, 0.5, 1, 1.5, 2, 2.5, 3\}$. Table S23 reports headline points from this sweep, while the corresponding figure provides the full landscape visualization. The $\alpha = 0$ point recovers the performance on the training mask, while $\alpha = 1$ represents a transition to a completely fresh device instance.

Table S23: D2D loss-landscape interpolation for canonical Standard vs Ensemble HAT ($NL = 1.0$).

Model	$\alpha = 0$ (train mask)	$\alpha = 1$ (fresh)	$\alpha = 3$ (extrapolated)
Standard HAT	94.67%	10.00%	9.55%
Ensemble HAT	91.25%	88.39%	27.06%

Table S23 and Figure S14 show that Ensemble HAT maintains 88.39% on fresh masks, whereas Standard HAT collapses to chance level (10.00%). At $\alpha = 3$, the sweep is deliberately extrapolated beyond the deployment distribution; the remaining 17.5 pp advantage over Standard HAT is therefore a stress-test indicator rather than a practical operating target. The primary evidence is the $\alpha = 1$ fresh-mask point.

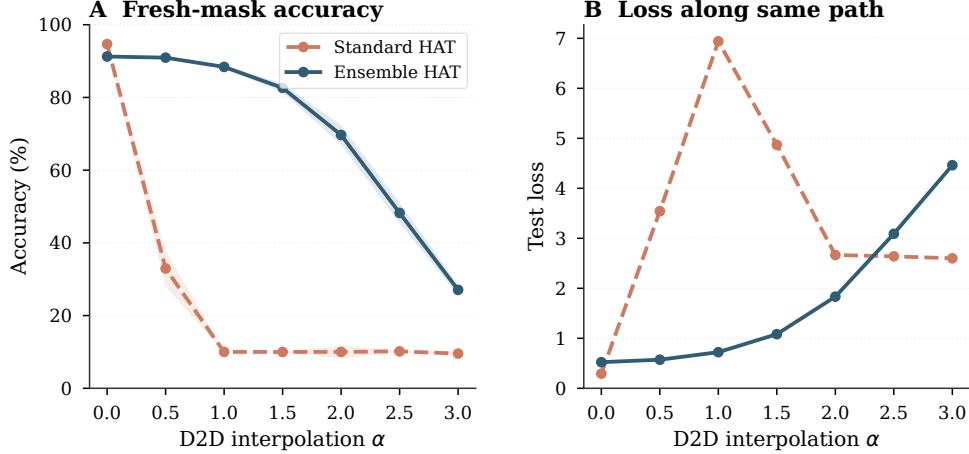


Figure S14: D2D-direction loss-landscape interpolation (E2). Standard HAT collapses when moving from the training mask to a fresh mask; Ensemble HAT remains accurate at the fresh-mask point. Shading denotes ± 1 standard deviation across three fresh masks; the $\alpha > 1$ region is an extrapolation beyond the training distribution.

S-M.2 Per-layer D2D sensitivity (E4)

We next examine layer-wise sensitivity by perturbing exactly one analog layer at a time with a fixed $\sigma_{\text{D2D}} = 10\%$ mask while zeroing all other D2D buffers. Rather than treating the resulting rank order as a localization proof, we use it as a diagnostic for where mismatch perturbations are most visible after Ensemble HAT has already controlled full-network fresh-instance transfer.

Table S24: Top-5 per-layer D2D sensitivity in canonical Ensemble HAT ($NL = 1.0$).

Rank	Layer	Group	Drop (pp)
1	stages.2.blocks.4.mlp.fc2	mlp	1.38
2	patch_embed.conv1.conv	patch_embed	1.18
3	stages.2.blocks.1.mlp.fc1	mlp	0.31
4	stages.2.blocks.5.mlp.fc2	mlp	0.29
5	stages.2.blocks.2.mlp.fc1	mlp	0.28

Four of the five most sensitive layers are MLP layers under the earlier source-domain diagnostic, while the patch-embedding convolution (rank 2) is also critical, consistent with its role as the first analog operation in the inference path. These sensitivity ranks are retained as mechanistic context, not as a main-text claim that severe-NL failure localizes primarily to the MLP path under the revised gradient-scaling recipe.

S-M.3 Full-parameter-space Hessian spectra (E1)

To characterize the local curvature of the loss surface, we approximate the Hessian eigenspectrum of the training loss over analog parameters using Lanczos iteration with Hessian-vector products (HVPs). The top absolute Ritz eigenvalue (Table S25) provides a proxy for the sharpest curvature direction in the parameter space.

Standard HAT exhibits a lower top eigenvalue (23.28) than Ensemble HAT (221.30) at the canonical $NL = 1.0$ regime. This is intentionally reported as a negative control: global Hessian flatness is not the relevant robustness metric for fresh hardware transfer. Standard HAT is globally flatter around its fixed training mask but still collapses when the D2D mask is replaced (Table S23). The better-aligned metric is D2D-directional sensitivity, captured directly by the interpolation in Fig. S14. The $NL = 2.0$ severe-NL checkpoints show much larger absolute top eigenvalues (1.76×10^3 – 3.01×10^4), consistent with increased loss-surface instability under

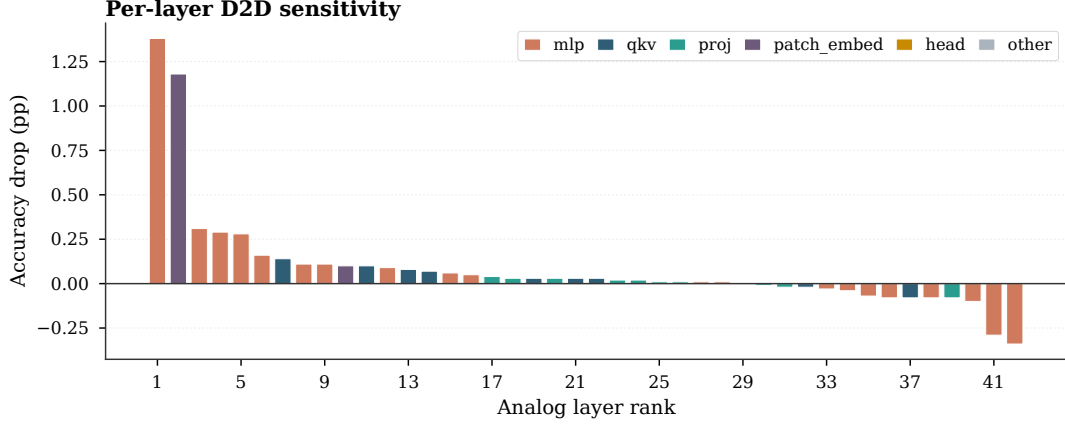


Figure S15: Per-layer D2D mismatch sensitivity (E4). Bars show accuracy drop when perturbing one analog layer at a time with a fixed $\sigma_{\text{D2D}} = 10\%$ mask; colors denote layer group.

Table S25: Top-1 absolute Hessian eigenvalue by checkpoint.

Checkpoint	Top-1 abs Ritz eigenvalue
Canonical Ensemble HAT ($NL = 1.0$)	221.30
Canonical Standard HAT ($NL = 1.0$)	23.28
M1 Standard ($NL = 2.0$)	30,058.40
M2 Ensemble ($NL = 2.0$)	5,705.60
M3 Proportional ($NL = 2.0$)	1,764.97

strong write nonlinearity.

Limitation. The Lanczos approximation uses a fixed eval batch of 32 samples. For 4.73×10^6 analog parameters, this may underestimate tail eigenvalues. The ranking across checkpoints is nevertheless informative.

S-M.4 Cross-checkpoint representational similarity (E3)

Linear Centered Kernel Alignment (CKA) across the M-series checkpoints (42 common analog layers) yields an aggregate off-diagonal CKA of 0.455. We use this value as an internal comparison only: random-initialized models would be expected to have much lower similarity, while same-architecture checkpoints trained under nearly identical recipes often show substantially higher CKA. The observed intermediate value supports a cautious conclusion that comparable $\sim 80\text{--}82\%$ severe-NL recovery can be reached through partially distinct representations, rather than through a single converged feature geometry.

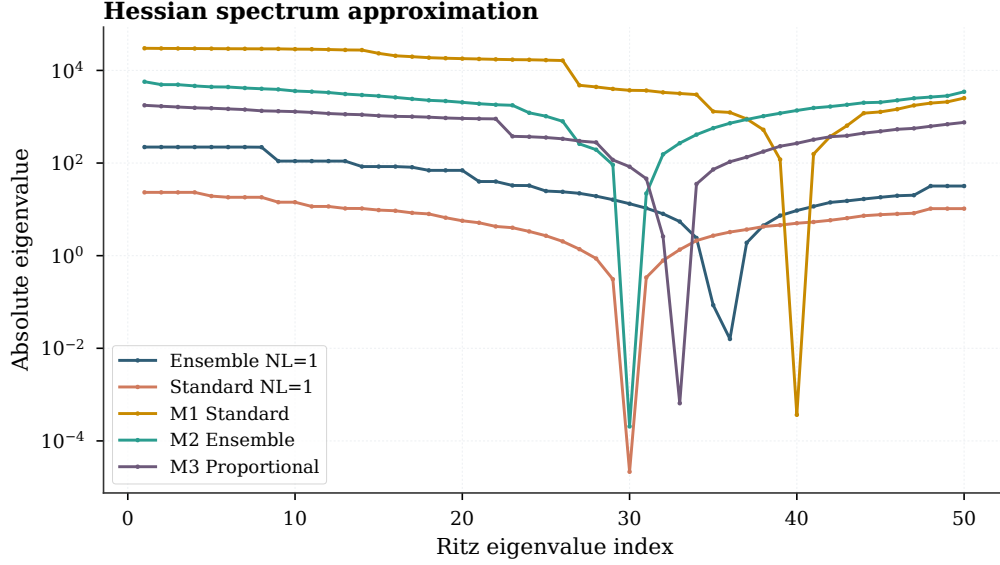


Figure S16: Hessian eigenspectrum approximation (E1). Lanczos Ritz eigenvalues over analog parameters show that mask-specific flatness does not predict fresh-instance transfer; severe-NL checkpoints have much larger eigenvalues.

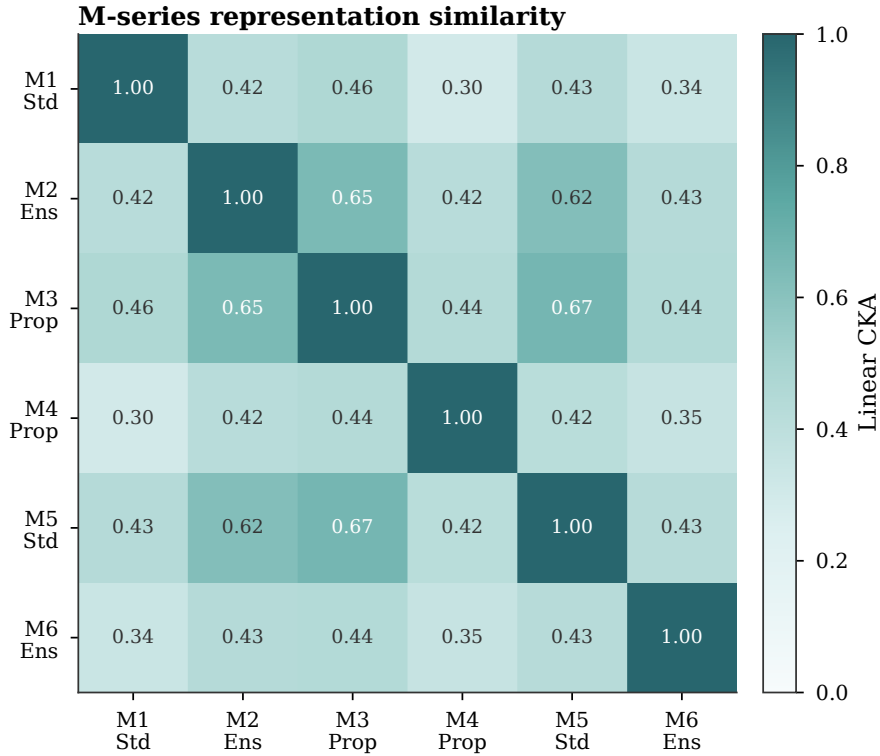


Figure S17: Cross-checkpoint linear CKA matrix (E3). Off-diagonal CKA shows that M-series checkpoints reach comparable accuracy through partially distinct internal representations.

S-M.5 Checkpoint averaging ablation (E5)

Naive parameter-space averaging of two Standard HAT checkpoints (M1 seed 123 and M5 seed 456) yields $10.00 \pm 0.00\%$ on fresh instances—identical to chance level. Thus, the robustness of Ensemble HAT should not be interpreted as ordinary checkpoint smoothing: the key ingredient is exposure to the mismatch distribution during optimization.

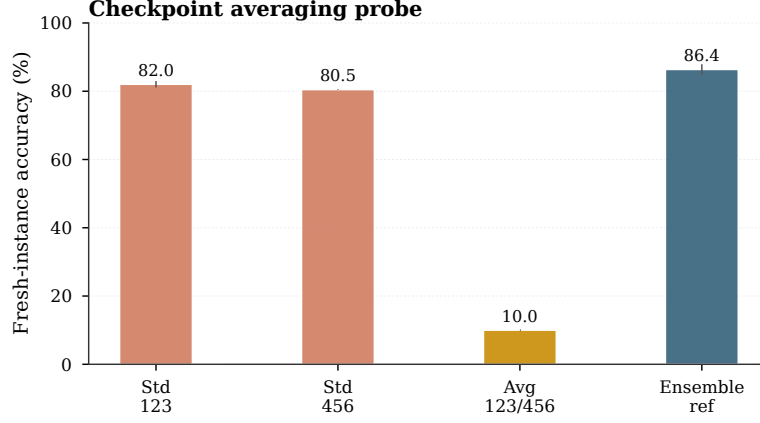


Figure S18: Checkpoint averaging ablation (E5). Naively averaging two Standard-HAT checkpoints remains at chance, so per-epoch D2D resampling is not equivalent to parameter-space smoothing.

Supplementary Note S-HW: Hardware Calibration Pipeline

This note describes the ingest pipeline for substituting measured device statistics into the simulation framework. To ensure that the high-level system conclusions can be validated against physical hardware, the pipeline is designed to accept four distinct categories of measured data. These inputs are then processed to produce a standardized JSON device profile, allowing the framework to serve as a bridge between fabricated materials and neural-network performance evaluation.

S-HW.1 Data categories

1. Pulsed programming curves: Conductance-vs-pulse measurements for LTP and LTD trajectories. The auto-fitter extracts $G_{\min}$, $G_{\max}$, N_{LTP} , and N_{LTD} via a behavioral exponential model.
2. Cycle-to-cycle statistics: Repeated write-verify measurements at identical target states, from which σ_{C2C} is extracted.
3. Device-to-device mismatch: Conductance distributions across multiple devices in a fabricated array, from which σ_{D2D} is extracted.
4. Retention decay: Long-term conductance-vs-time measurements, from which τ_1 , τ_2 , and A_0 are fitted via dual-exponential regression.

S-HW.2 Current status

At the time of submission, measured-device data from the fabricated array was not yet available. All reported results therefore use literature-derived profiles. When measured statistics become available, the same workflow can be rerun with the measured profile substituted for the literature prior, preserving comparability across all reported experiments.

Supplementary Note S-Tooling: Tooling Positioning in the Analog CIM Ecosystem

This note situates the present framework within the broader landscape of analog compute-in-memory (CIM) simulation and training tools. Given the rapid evolution of this field, it is critical to perform honest comparative positioning. Our work is intended to be complementary to existing, mature toolkits by addressing specific architectural needs for transformer-based

vision models that are not currently covered by standard libraries. We also acknowledge and cite the conceptual lineage that informed our implementation choices.

S-T.1 Quantitative cross-check

We performed a controlled cross-comparison on a 1000-image deterministic CIFAR-10 subset:

- Clean baseline: Our framework reports 86.2%; CrossSim reports 83.7% at 8-bit ADC.
- Noisy 3-run Monte Carlo ($\sigma_{D2D} = 5\%$): We report $81.63 \pm 0.56\%$; CrossSim reports $67.20 \pm 2.67\%$.

The 14.4 pp divergence should be interpreted as a convention mismatch rather than as an apples-to-apples benchmark. Our framework uses multiplicative D2D mismatch relative to each mapped conductance range, whereas the CrossSim configuration used here applies an additive programming-error/read-noise model in absolute conductance units. These two noise laws are not mathematically equivalent unless the conductance range and scaling conventions are explicitly reparameterized to match. We therefore use this experiment only as a sanity check that both simulators reproduce the same qualitative vulnerability to analog mismatch; we do not claim numerical superiority over CrossSim from this subset comparison. CrossSim is also substantially slower per evaluation for full parameter sweeps, so the present framework is used for design-space exploration while CrossSim remains a mature reference simulator for inorganic-array studies.

S-T.2 Conceptual ancestry

We acknowledge AIHWKit (Rasch et al. 2023) as the conceptual ancestor of the train-with-differentiable-surrogate / eval-with-ADC discipline and the per-epoch noise resampling primitives used in the present framework.

S-T.3 AIHWKit head-to-head attempt

We attempted a direct head-to-head replication of the canonical Tiny-ViT V4 checkpoint in AIHWKit v0.9.1. Conversion failed with **ValueError: Only one group is supported**, because Tiny-ViT’s patch-embedding and stage-0 blocks employ grouped convolutions (depthwise-separable filters) that AIHWKit’s **AnalogConv2d** does not currently map. A ResNet-18 equivalent (no grouped convolutions) converts successfully, but that architecture is outside the present study’s scope. This limitation underscores why a custom framework was necessary for Vision-Transformer CIM evaluation: contemporary open-source analog-training toolkits are optimized for standard convolutions and fully-connected layers, not for the heterogeneous operator mix (grouped conv, attention, MLP) found in modern ViTs.

Supplementary Note S-Reproducibility

To facilitate the reproduction and extension of the results presented in this study, the full source code for the simulation framework and the generated device profiles will be made available. This includes the core analog-CIM operator implementations, the hardware-aware training (HAT) scripts, and the evaluation pipelines for various physical non-idealities. All materials are hosted at an anonymous repository to be disclosed upon acceptance.

2.7 Class-Distribution Analysis of the Fresh-Instance Collapse Mechanism

Figure S19 compares per-class prediction distributions for the canonical fixed-mask Standard HAT checkpoint and the canonical Ensemble HAT checkpoint under fresh D2D instances at $NL = 1.0$.

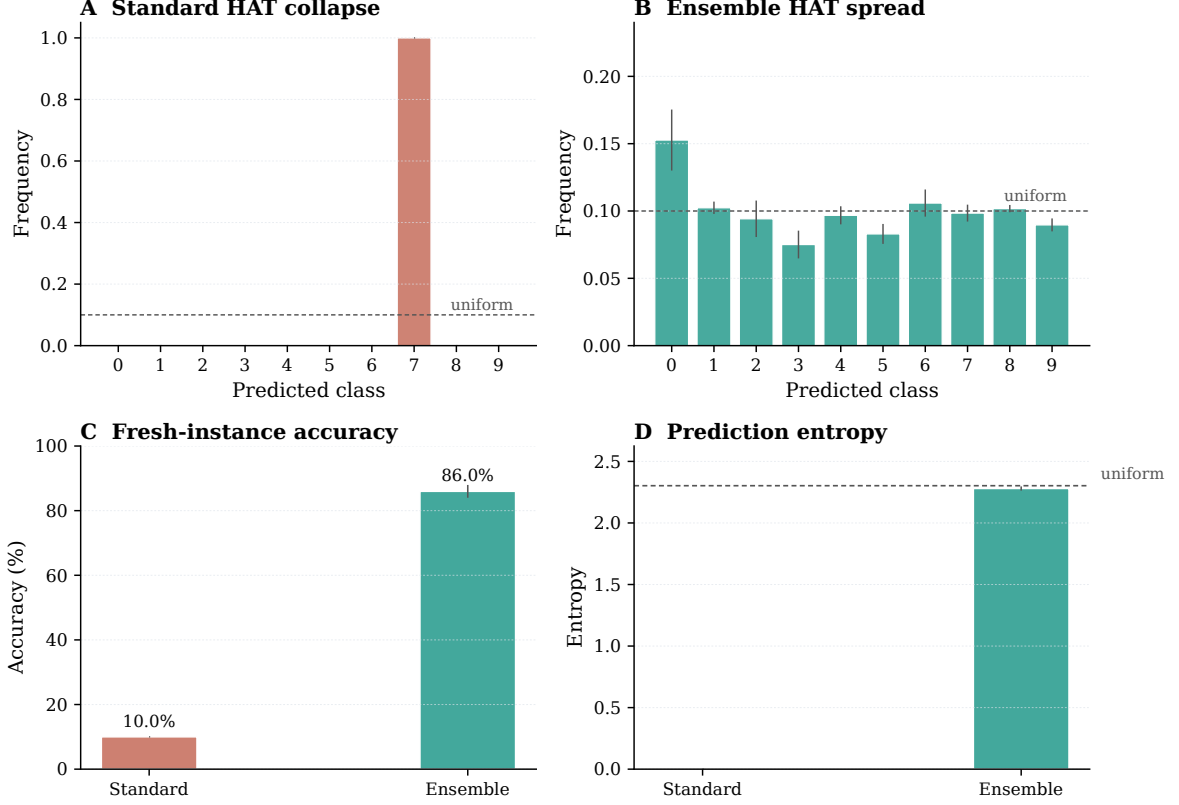


Figure S19: Class-distribution diagnostic. Fixed-mask Standard HAT predicts one class on fresh D2D instances; Ensemble HAT preserves near-uniform prediction entropy under the same protocol.

2.8 Full Local PCM Precision-Ladder Per-Seed Tables

The following tables detail the raw performance metrics used for the PCM precision ladder. After the protocol audit on 2026-05-09, the 4-bit and 8-bit entries remain canonical, whereas the original 6-bit row was retired because its training-time evaluation used `enable_during_test=False`. The corrected 6-bit table below uses retrained new-protocol checkpoints with `enable_during_test=True`. Thus, 6-bit is reported as a D2D-sensitive transition regime rather than as a frontier optimum.

We also tested a strict 5-bit PCMPresetUnitCell pilot. The best documented strict seed reached only $63.44 \pm 0.07\%$ fresh accuracy with a 2.52 pp one-day drift drop, below the 70% continuation gate. We therefore treat 5-bit as non-frontier and omit it from the canonical precision ladder.

Table S26: 8-bit PCM UnitCell. 3-seed Mean: 77.64%, Fresh: 77.60%, Drop: 0.04 pp.

Seed	Best Test	Fresh Mean $\pm$ Std	Drift 1d	Δ Drift (1d)
123	77.00%	77.00 $\pm$ 0.05%	77.01%	-0.01 pp
456	78.36%	78.27 $\pm$ 0.05%	78.22%	0.05 pp
789	77.56%	77.52 $\pm$ 0.04%	77.49%	0.03 pp

Table S27: 6-bit PCM UnitCell, corrected new protocol. Four-seed fresh mean: 68.44% with 6.03 pp seed-to-seed standard deviation. The completed 24 h drift closure gives a mean drift drop of 0.04 pp, confirming that the corrected 6-bit regime is drift-flat but seed-sensitive.

Seed	Best Test	Fresh Mean $\pm$ Std	Drift 1d	Δ Drift (0s-1d)
123	68.51%	68.49 $\pm$ 0.03%	68.58%	-0.14 pp
456	62.60%	62.47 $\pm$ 0.06%	62.12%	0.29 pp
457	76.63%	76.69 $\pm$ 0.05%	76.68%	-0.05 pp
789	65.86%	66.13 $\pm$ 0.04%	66.07%	0.06 pp

Table S28: 4-bit PCM UnitCell. 3-seed Mean: 76.71%, Fresh: 76.68%, Drop: 4.01 pp.

Seed	Best Test	Fresh Mean $\pm$ Std	Drift 1d	Δ Drift (0s-1d)
123	76.74%	76.65 $\pm$ 0.04%	72.28%	4.40 pp
456	77.15%	77.07 $\pm$ 0.06%	72.18%	4.84 pp
789	76.23%	76.33 $\pm$ 0.04%	73.45%	2.78 pp

2.9 Retired 6-bit PCM Late-Recovery Diagnostic

The former late-recovery plot for 6-bit seed 456 is no longer included as an active supplementary figure. That diagnostic used an old-protocol checkpoint whose training-time evaluation disabled test-time noise injection; it therefore supported a protocol-specific recovery claim rather than a deployment-facing PCM result. Under the corrected protocol, seed 456 reaches 62.60% best test accuracy and $62.47 \pm 0.06\%$ fresh accuracy, while seed 457 reaches 76.63% best test accuracy and $76.69 \pm 0.05\%$ fresh accuracy. The appropriate conclusion is high seed sensitivity in the 6-bit transition regime, not a universal late-recovery effect.

2.10 Provenance and Checkpoint Tracking

To ensure rigorous reproducibility, all canonical results are mapped to their generating scripts and deterministic seeds. The 4-bit and 8-bit PCM aggregates use the original audited checkpoints. The 6-bit aggregate uses corrected new-protocol checkpoints with `modifier_enable_during_test=True`; seeds 456/457/789 have training histories, while seed 123 has fresh/drift evaluation files but no saved training history in the current archive.

Table S29: Provenance and Checkpoint Tracking.

Precision	Seed	Checkpoint Directory	Schedule
8-bit	123	r11d_5a_pcm_seed123	patience=0
8-bit	456	r11d_5a_pcm_seed456	patience=0
8-bit	789	r11d_5a_pcm_seed789	patience=0
6-bit	123	r11d_6bit_pcm_seed123	new protocol; history missing
6-bit	456	r11d_6bit_pcm_seed456	patience=10; best epoch 30
6-bit	457	r11d_6bit_pcm_seed457	patience=10; best epoch 100
6-bit	789	r11d_6bit_pcm_seed789	patience=10; best epoch 44
4-bit	123	r11d_7_pcm_4bit_seed123	patience=0
4-bit	456	r11d_7_pcm_4bit_seed456_clean	patience=0
4-bit	789	r11d_7_pcm_4bit_seed789	patience=0

2.11 Preliminary Remote Validation

Preliminary remote validation on DeiT/ViT suggests proportional noise-aware training preserves fresh-instance accuracy across transformer backbones, pending full multi-seed validation.

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